Sinusoids and Phasors

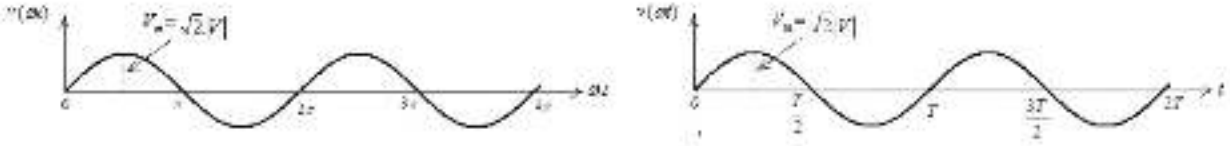
AC circuits have time varying sources, usually producing sinusoidal voltage and current waveforms, also called sinusoids.
A sinusoidal voltage has the form:

$$v(t) = V_m.\sin(\omega t) \qquad (1)$$

or, $v(t) = V_m.\cos(\omega t)$

Here, V_m represents the maximum value (amplitude) and ω is the angular frequency.

See the figures below for the waveform $v(t) = V_m.\sin(\omega t)$.



Note that small letters are used to represent instantaneous values.

This voltage waveform can be written as:

$$v(t) = V_m.\sin(\omega t) \text{ or } v(t) = V_m.\cos(\omega t - \frac{\pi}{2})$$

Note that that the sinusoid repeats itself every T seconds; thus, T is called the *period* of the sinusoid.
From the previous two plots, we observe that: $\omega T = 2\pi$ or:

$$T = \frac{2\pi}{\omega} \qquad (2)$$

Note also that $v(t) = v(t+T)$ or the voltage waveform is a periodic function of time with a period equals T.

As mentioned, the *period* T of the periodic function is the time of one complete cycle. The reciprocal of this quantity is the number of cycles per second, known as the *cyclic frequency* f of the sinusoid. Thus,

$$f = \frac{1}{T} \qquad (3)$$

From (2) and (3), $\frac{1}{f} = \frac{2\pi}{\omega}$, or:

$$\omega = 2\pi f \qquad (5)$$

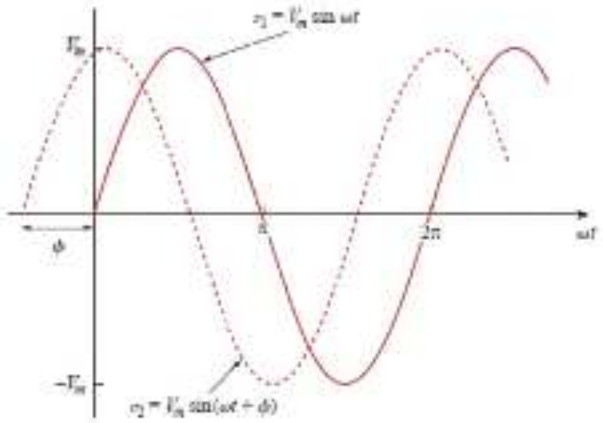
Note that ω is measured in rad/s, while f is in hertz (Hz).

Consider the more general expression for the sinusoid:

$$v(t) = V_m.\sin(\omega t + \phi) \qquad (6)$$

Here ϕ is the phase shift.

The figure below shows the two sinusoids $v_1(t) = V_m.\sin(\omega t)$ and $v_2(t) = V_m.\sin(\omega t + \phi)$



Note that the starting point of $v_2(t)$ occurs first in time. Therefore, we say that $v_2(t)$ *leads* $v_1(t)$ by ϕ, or that $v_1(t)$ *lags* $v_2(t)$ by ϕ.

If $\phi \neq 0$ we also say that $v_1(t)$ and $v_2(t)$ are *out of phase*. If $\phi = 0$ we say that $v_1(t)$ and $v_2(t)$ are *in phase*; they reach their minima and maxima at exactly the same time.

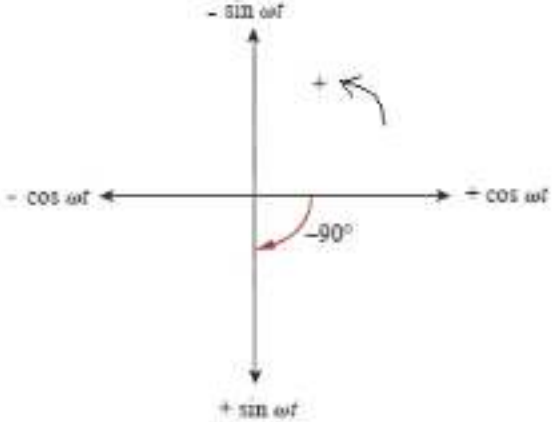
Note that we can compare $v_1(t)$ and $v_2(t)$ in this manner because they operate at the same frequency.

Note that a sinusoid can be expressed in either sine or cosine form. When comparing two sinusoids, it is expedient to express both as either sine or cosine functions with positive amplitudes. This is achieved by using the following trigonometric identities:

$$\sin(A+B) = \sin A.\cos B + \cos A.\sin B$$

$$\cos(A+B) = \cos A.\cos B - \sin A.\sin B$$

A graphical approach can also be used to relate or compare sinusoids. Consider the figure below.



The horizontal axis represents the magnitude of cosine, while the vertical axis (pointing down) denotes the magnitude of sine. Angles are measured positively counterclockwise from the horizontal, as usual in the polar coordinates.

This graphical technique can be used to relate two sinusoids. For example:

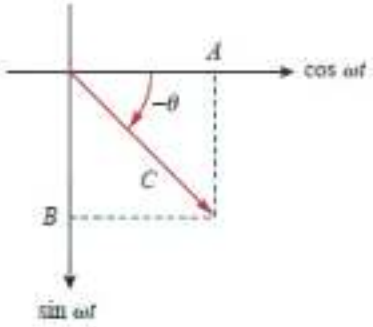
$\cos(\omega t - 90^0) = \sin(\omega t)$

$\sin(\omega t - 90^0) = -\cos(\omega t)$

$\cos(\omega t + 180^0) = -\cos(\omega t)$

$\sin(\omega t + 180^0) = -\sin(\omega t)$

The graphical technique can also be used to add two sinusoids of the same frequency when one is in sine form and the other is in cosine form. To add $A\cos\omega t$ and $B\sin(\omega t)$, from the figure below:



$A\cos\omega t + B\sin\omega t = C\cos(\omega t - \theta) = C\sin(\omega t + \theta)$

Where:

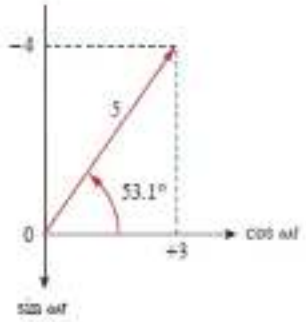
$C = \sqrt{A^2 + B^2}$ and $\theta = \tan^{-1}\left(\frac{B}{A}\right)$

Example:

Find $3\cos\omega t - 4\sin\omega t$

Solution:

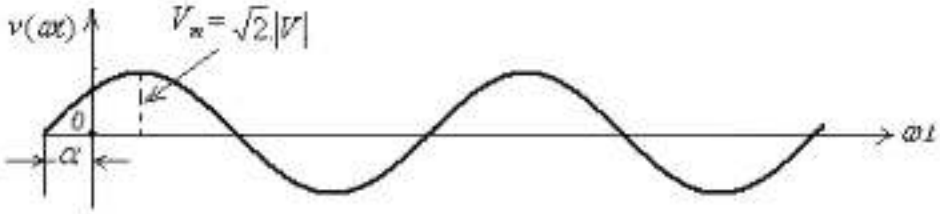
Consider the figure below.



Thus:

$3\cos\omega t - 4\sin\omega t = 5\cos(\omega t + 53.1^\circ)$

Consider the following voltage waveform.

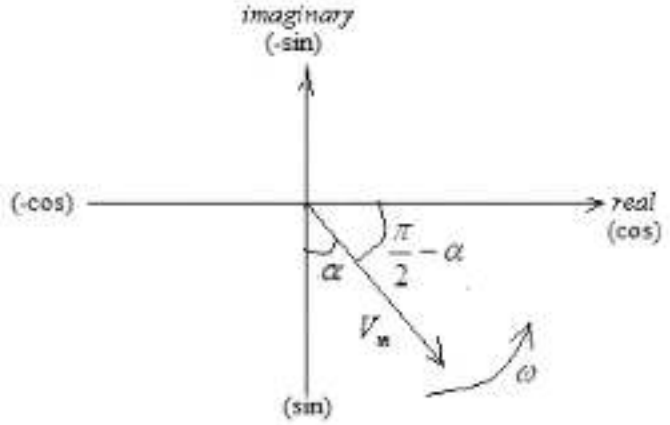


This waveform as a function of time can be written as:

$v(t) = V_m.\sin(\omega t + \alpha)$, or

$$v(t) = V_m.\cos\left(\frac{\pi}{2} - \omega t - \alpha\right) = V_m \cos\left[\omega t - \left(\frac{\pi}{2} - \alpha\right)\right]$$

This waveform can also be represented as a phasor as shown below.



Note that the vector V is rotating in the positive direction (anticlockwise) with an angular speed equals to ω. The instantaneous value of this vector as a function of time can be found as the projection of this vector on the real axis.

note that at $\omega t = 0$, from the time diagram $v(0) = V_m.\sin(\alpha) = V_m.\cos\left(\frac{\pi}{2} - \alpha\right)$, which also can be found from the phasor diagram.

at $\omega t = \frac{\pi}{2} - \alpha$, the instantaneous value of the voltage is V_m and the phasor will coincide with the horizontal axis.

Note also from the phasor representation of the vector V, this vector can be written also as:

$V = |V| \angle \alpha$ or as $V = |V| e^{j\alpha}$ these are called polar forms, where $|V|$ is the length of the phasor.

Also this phasor can be represented as: $V = |V|(\cos\alpha + j\sin\alpha)$, this called a Cartesian form and is produced from Euler's formula, as: $e^{j\alpha} = \cos\alpha + j\sin\alpha$.

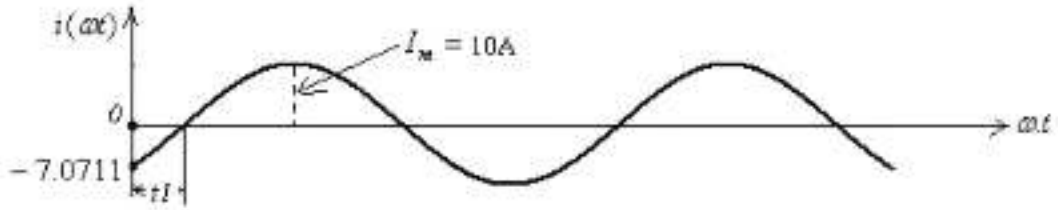
The instantaneous value of this vector as a function of time can be found as the projection of this vector on the real axis.

Note: to change from sine to cosine expressions, the following identities can also be used:

$\cos(x) = \sin(\frac{\pi}{2} - x)$ and $\sin(x) = \cos(\frac{\pi}{2} - x)$. Also $\cos(-x) = \cos(x)$ and $\sin(-x) = -\sin(x)$

Example:

for the sinusoidal current waveform shown.



a) express the instantaneous value of the current as a sin and as a cos function of ωt, i.e., find the expression of $i(\omega t)$

b) draw the phasor diagram of this current.

c) express this current waveform in polar and in Cartesian forms.

d) if the frequency of this waveform is 50Hz., find the period in seconds.

e) find the time t_1.

Solution:

a) from the figure, the maximum value is 10A

The expression of the current is $i(\omega t) = 10.\sin(\omega t + \alpha)$

at $\omega t = 0$, $i = -7.07110$, or:

$-7.0711 = 10.\sin(\alpha)$

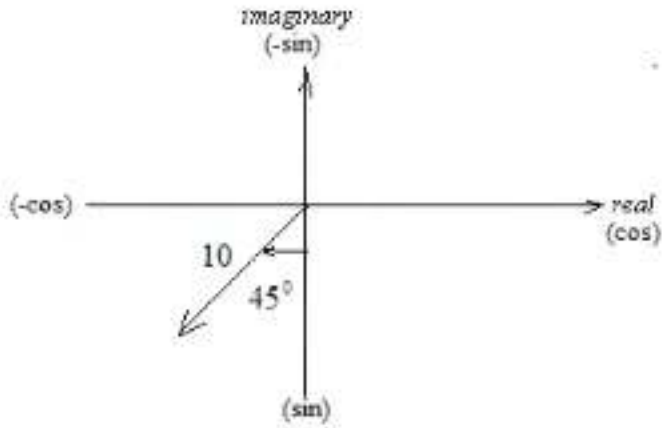
from it: $\alpha = -\frac{\pi}{4}$

or the expression of the current is: $i(\omega t) = 10.\sin(\omega t - \frac{\pi}{4})$.

This waveform also can be written as:

$$i(\omega t) = 10.\cos(\frac{\pi}{2} - \omega t + \frac{\pi}{4}) = 10.\cos(\frac{3\pi}{4} - \omega t) = 10.\cos(\omega t - \frac{3\pi}{4}) = 10.\cos(\omega t - 135^0)$$

b) The phasor diagram is as follow:



Note that the length of the current phasor is 10

c) in a polar form $I = 10\angle -135^0$ or $I = 10.e^{-j\frac{3\pi}{4}}$
in Cartizian form $I = 10.\left[\cos(-135^0) + j\sin(-135^0)\right] = -7.0711 - j7.0711$

d) $T = \frac{1}{f} = \frac{1}{50} = 20ms.$

e) $t_1 = \frac{20x\frac{\pi}{4}}{2\pi} = 2.5ms$

Note: the length of the phasor in some phasor diagrams can be the rms (maximum divided $\sqrt{2}$) value of the waveform. This has to be kept in mind during calculations.

Derivation and integration of phasors

If the volage waveform $v(t)$ is represented by the phasor $V = V_m e^{j\phi}$, thus its derivative will be:

$$\frac{d}{dt}v(t) = j\omega\left(V_m e^{j\phi}\right) = j\omega V$$

Or derivation in the time domain is equivalent to multiplication by $j\omega$ in the phasor domain.

In the same way:

$$\int v(t)dt = V_m \int e^{j\phi} dt = \frac{1}{j\omega} j\omega$$

Or integration in the time domain is equivalent to dividing by $j\omega$ in the phasor domain.

Note:
When a waveform is represented as a phasor, the rules of complex numbers are used. To sum or subtract two complex numbers use the cartesian form. While to divide or multiply numbers use polar forms.

Examples

Example 1:

Find the amplitude, phase, frequency and the period of the sinusoid:

$v(t) = 100.\cos\left(70t + 30^0\right)$

Solution:

The amplitude is $V_m = 100V$

The phase is $\phi = 30^0$

the angular frequency is $\omega = 70 rad/s$ and the frequency is $f = \frac{\omega}{2\pi} = \frac{70}{2\pi} = \frac{35}{\pi} Hz.$

the period is $T = \frac{1}{f} = \frac{\pi}{35} s.$

Example 2:

Calculate the phase angle between $v_1(t) = 10\cos(\omega t - 30^0)$ and $v_2(t) = 30\sin(\omega t + 45^0)$

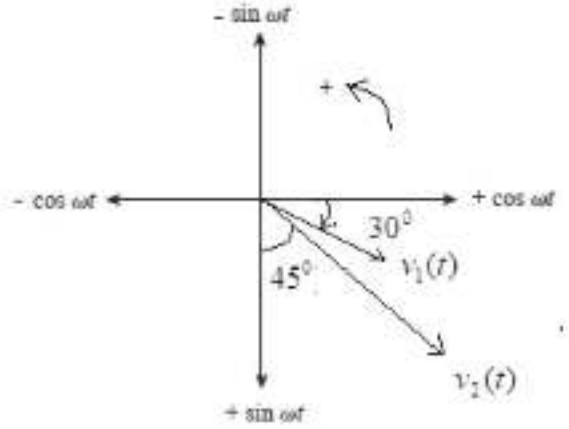
Solution:

Transform $v_2(t)$ into cosine functions, thus:

$$v_2(t) = 30\sin(\omega t + 45^0) = 30\cos\left[90^0 - \left(\omega t + 45^0\right)\right] = 30\cos\left(45^0 - \omega t\right) = 30\cos\left(\omega t - 45^0\right)$$

As $v_1(t) = 10\cos(\omega t - 30^0)$ and $v_2(t) = 30\cos\left(\omega t - 45^0\right)$, thus the phase angle between the two waveforms is 15^0 and $v_2(t)$ is leading.

The same answer can be found using the graphical method as shown below.



Example 3:

Evaluate the complex number $\sqrt{\dfrac{8\angle -60^0 + (3+j4)}{(5+j6)(3-j5)^*}}$

Solution:

$$8\angle -60^0 = 8\left[\cos(-60^0) + j\sin(-60^0)\right] = 4 - j4\sqrt{3}$$

$$8\angle -60^0 + (3+j4) = 4 - j4\sqrt{3} + (3+j4) = 7 - j4(\sqrt{3}-1) = 7 - j2.9282 = 7.5878\angle -22.7^0$$

$$(3-j5)^* = 3 + j5$$

$$(5+j6)(3-j5)^* = (5+j6)(3+j5) = (5x3 - 6x5) + j(6x3 + 5x5) = -15 + j43 = 45.5412\angle 109.23^0$$

$$\frac{8\angle -60^0 + (3+j4)}{(5+j6)(3-j5)^*} = \frac{7.5878\angle -22.7}{45.5412\angle 109.23^0} = 0.1666\angle -131.93^0$$

Thus:

$$\sqrt{\frac{8\angle -60^0 + (3+j4)}{(5+j6)(3-j5)^*}} = \sqrt{0.1666\angle -131.93^0} = \sqrt{0.1666}\angle -\frac{131.93^0}{2} = 0.4082\angle -65.965^0$$

$$= 0.1663 - j0.3728$$

Example 4:

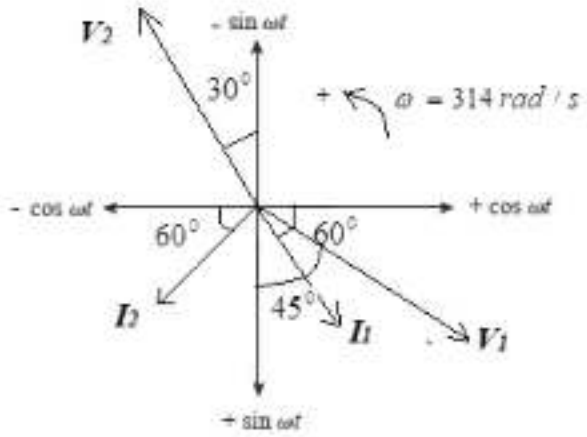
Transform these sinusoids to phasors:

a) $i_1(t) = 5\cos(314t - 60^0)$ b) $v_1(t) = 10\sin(314t + 45^0)$

c) $i_2(t) = -5\cos(314t + 60^0)$ d) $v_2(t) = -10\sin(314t - 30^0)$

Solution:

The easiest method is to use the graphical diagram as shown below.



From this diagram, it can be seen that:

a) $I_1 = 5\angle -60^0$ b) $V_1 = 10\angle -45^0$

c) $I_2 = 5\angle 240^0$ d) $V_2 = 10\angle 120^0$

note: here the length of the phasor represents the peak value of the variable. In some diagrams the length of the phasor represents the RMS value of the variable.

Example 5:

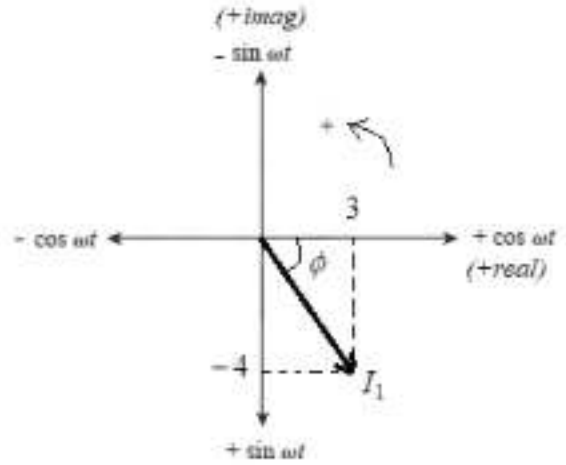
Find the sinusoids represented by these phasors:

a) $I_1 = 3 - j4$

b) $V_1 = j10e^{-j30^0}$

Solution:

a) the following diagram shows the phasor I_1



From this diagram:

$$|I_{s1}| = \sqrt{3^2 + 4^2} = 5, \quad \phi = -\tan^{-1}\left(\frac{4}{3}\right) = -53.13^0$$

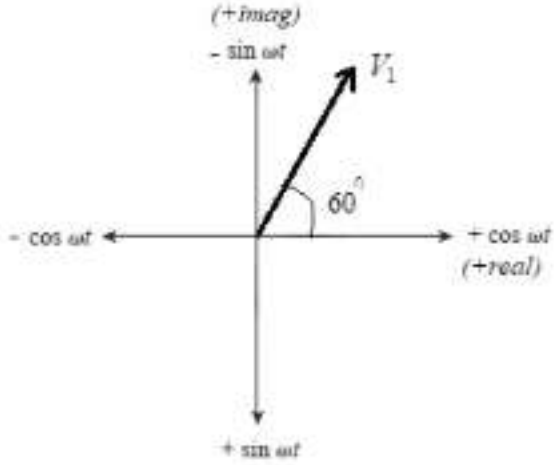
Or:

$I_1 = 3 - j4 = 5\angle -53.13^0$ and:

$i_1(t) = 5\cos(\omega t - 53.13^0) = 5\sin(\omega t + 36.87^0)$

b) $V_1 = j10e^{j30^0} = \left(e^{j90^0}\right)\left(10e^{j30^0}\right) = 10e^{j60^0}$, note that $j = e^{j90^0}$

the phasor V_1 is shown in the following diagram.



And:

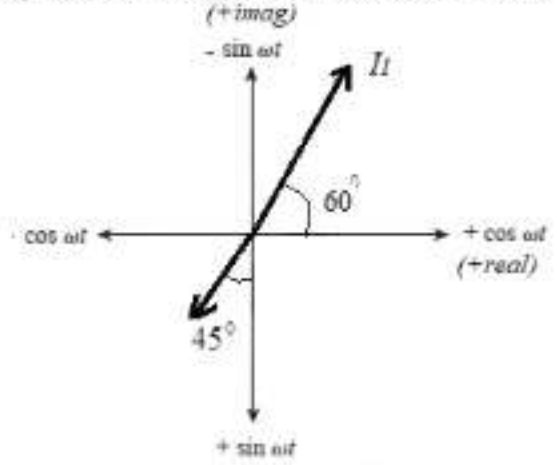
$v_1(t) = 10\cos(\omega t + 60^0) = 10\sin(\omega t + 150^0)$

Example 6:

Find the sum of the two sinusoids: $i_1(t) = 10\cos(\omega t + 60^0)$ and $i_2(t) = 5\sin(\omega t - 45^0)$

Solution:

The figure below shows the phasors of these two current waveforms.



From this diagram $I_1 = 10\angle 60^0$ and $i_2(t) = 5\angle -135^0$, thus:

$$I = I_1 + I_2 = 10\angle 60^0 + 5\angle -135^0 = 10\left[\cos(60^0) + j\sin(60^0)\right] + 5\left[\cos(-135^0) + j\sin(-135^0)\right]$$

$$= 10\left[\frac{1}{2} + j\frac{\sqrt{3}}{2}\right] + 5\left[-\frac{1}{\sqrt{2}} - j\frac{1}{\sqrt{2}}\right] = \left(5 - \frac{5}{\sqrt{2}}\right) + j\left(5\sqrt{3} - \frac{5}{\sqrt{2}}\right) = 1.4644 - j5.1247 = 5.3298\angle 74.05^\circ$$

Or:

$$i(t) = i_1(t) + i_2(t) = 5.3298\cos\left(\omega t + 74.05^0\right)$$

Example 6:

Using the phasor approach, determine the current in a circuit described by the integrodifferential equation: $4i + 8\int i dt - 3\frac{di}{dt} - 50\cos(2t + 75^0)$

Solution:

The phasor form of this equation is:

$4I + \frac{8}{j\omega}I - 3(j\omega)I = 50\angle 75^0$, where $\omega = 2$, thus this equation becomes:

$$4I - 4jI - 6jI = 50\angle 75^0$$

$$(4 - 10j)I = 50\angle 75^0$$

$$I = \frac{50\angle 75^0}{4 - 10j} = \frac{50\angle 75^0}{10.77\angle -68.2^0} = 4.642\angle 143.2^0$$

Or:

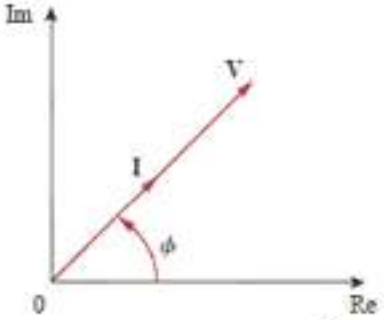
$$i(t) = 4.642\cos\left(2t + 143.2^0\right)$$

Phasor Relations for Circuit Elements

By circuit elements we mean resistors (R), inductors (L) and capacitors (C).

Resistor: If the current through a resistor R is $i(t) = I_m \cos(\omega t + \phi)$, the phasor of which is $I = I_m \angle \phi$, the voltage across it is given by Ohm's law, or $v(t) = R.i(t) = R.I_m \cos(\omega t + \phi)$, the phasor of which is $V = R.I_m \angle \phi$, or $V = R.I$.

Or, the voltage between the terminals of a resistor and the current through it are in phase, as illustrated in the phasor diagram shown below.



Inductor: If the current through an inductor L is $i(t) = I_m \cos(\omega t + \phi)$, the phasor of which is $I = I_m \angle \phi$, the voltage across it is given by:

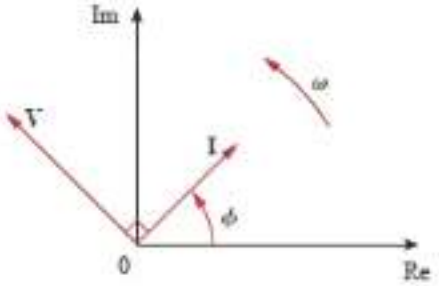
$v(t) = L.\dfrac{di(t)}{dt} = -\omega L I_m \sin(\omega t + \phi)$, which can be written as $v(t) = \omega L I_m \cos(\omega t + 90^0 + \phi)$, the phasor of which is $V = \omega L I_m \angle(90^0 + \phi)$. Or the current and voltage phasors in exponential forms are:

$I = I_m e^{j\phi}$ and $V = \omega L I_m e^{j(90^0+\phi)} = \omega L I_m e^{j90^0} e^{j\phi}$,

As $e^{j90^0} = j$, thus:

$V = j\omega L I_m e^{j\phi} = j\omega L.I$ (1)

Or the voltage has a magnitude of $\omega L I_m$ and a phase of $\phi + 90^0$. The voltage and the current are 90^0 out of phase. Specifically, the current lags the voltage by 90^0, as shown below.



Equation (1) can be written as:

$V = j\omega L.I = X_L.I$ or the reactance on an inductor is given by:

$X_L = j\omega L = j(2\pi f)L$

Capacitor: If the voltage across a capacitor is $v(t) = V_m \cos(\omega t + \phi)$, the phasor of which is $V = V_m \angle \phi$, the current through the capacitor is given by:

$i(t) = C\dfrac{dv(t)}{dt} = -\omega C V_m \sin(\omega t + \phi)$, which can be written as $i(t) = \omega C V_m \cos(\omega t + 90^0 + \phi)$, the phasor of which is $I = \omega C V_m \angle(90^0 + \phi)$. Or the voltage and the current phasors in exponential forms are:

$V = V_m e^{j\phi}$ and $I = \omega C V_m e^{j(90^0 + \phi)} = \omega C V_m e^{j90^0} e^{j\phi}$.

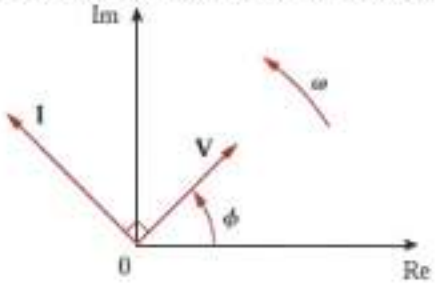
As $e^{j90^0} = j$, thus:

$I = j\omega C V_m e^{j\phi} = j\omega C V$

Or:

$$V = \frac{I}{j\omega C} = -j\frac{1}{\omega C} I \qquad (2)$$

Or the current has a magnitude of $\omega C V_m$ and a phase of $\phi - 90^0$. The voltage and the current are 90^0 out of phase. Specifically, the current leads the voltage by 90^0, as shown below.



Equation (2) can be written as:

$V = \dfrac{I}{j\omega C} = -j\dfrac{1}{\omega C} I = X_C . I$ or the reactance on a capacitor is given by:

$$X_C = \frac{1}{j\omega C} = -j\frac{1}{\omega C} = -j\frac{1}{2\pi f C}$$

Example 1:

If voltage $v(t) = 20\cos(314t + 60^0)$ is applied to a $50\mu F$ capacitor, calculate the steady-state current through the capacitor.

Solution:

$$X_C = \frac{1}{j\omega C} = -j\frac{1}{314(50x10^{-6})} = -j63.69\Omega$$

$V = 20\angle 60^0 V$

$$I = \frac{V}{X_C} = \frac{20\angle 60^\circ}{63.69\angle -90^\circ} = 0.314\angle 150^\circ\ A$$

Or:

$$i(t) = 0.314\cos(314t + 150^\circ)A$$

Example 2:

If voltage $v(t) = 20\cos(314t + 60^\circ)$ is applied to a $50mH$ inductor, calculate the steady-state current through the inductor.

Solution:

$$X_L = j\omega L = j(314)(50\times10^{-3}) = j15.7\Omega$$

$$V = 20\angle 60^\circ V$$

$$I = \frac{V}{X_L} = \frac{20\angle 60^\circ}{15.7\angle 90^\circ} = 1.274\angle -30^\circ A$$

Or:

$$i(t) = 1.274\cos(314t - 30^\circ)A$$

Impedance and Admittance

The impedance Z of a circuit is the ratio of the phasor voltage V to the phasor current I, measured in ohms (Ω). It represents the opposition that the circuit exhibits to the flow of sinusoidal current.

Or:

$$Z = \frac{V}{I}$$

Note that the impedance is not a phasor, because it does not correspond to a sinusoidally varying quantity.

The electrical circuit in general can have resisters, inductors and capacitors and the impedance of a given circuit can be a complex number. The real part of the impedance is a resistor and the imaginary part is called a reactance.

For a series RL circuit, the impedance is:

$$Z = \frac{V}{I} = R + j\omega L = R + jX_L$$

Note that the real part of Z is the resistor R and the imaginary part is the reactance, $X_L = \omega L$.

For a series RC circuit, the impedance is:

$$Z = \frac{V}{I} = R + \frac{1}{j\omega C} = R - j\frac{1}{\omega C} = R - jX_C$$

Also for this circuit, the real part of Z is the resistor R and the imaginary part is the reactance

$$X_C = \frac{1}{\omega C}$$

Note that for the two extreme cases, when $\omega = 0$ (circuits with DC sources):

- the reactance of the inductor is $X_L = \omega L = 0$, or the inductor reacts as a short-circuit.

- the reactance of the capacitor is $X_C = \dfrac{1}{\omega C} = \infty$, or the capacitor reacts as an open-circuit.

when $\omega \to \infty$ (circuits with very high frequency sources):

- the reactance of the inductor is $X_L = \omega L \to \infty$, or the inductor reacts as an open-circuit.

- the reactance of the capacitor is $X_C = \dfrac{1}{\omega C} \to 0$, or the capacitor reacts as a short-circuit.

As the impedance is a complex number, it can be represented in a polar form, or:

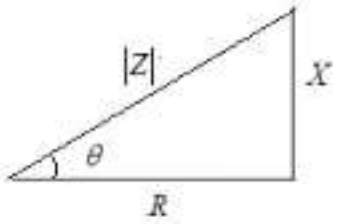
$Z = R + jX = |Z| \angle \theta$, where:

$|Z| = \sqrt{R^2 + X^2}$ and $\theta = \tan^{-1}\left(\dfrac{X}{R}\right)$

And:

$R = |Z| \cos\theta, \quad X = |Z| \sin\theta$

See the impedance triangle shown below.



the reciprocal of the impedance is known as admittance Y and measure in Siemens (S) or mho ($\mho$) , or:

$$Y = \frac{1}{Z} = \frac{I}{V} = G + jB$$

Note that G is called the admittance and B is called the susceptance.

Note also that:

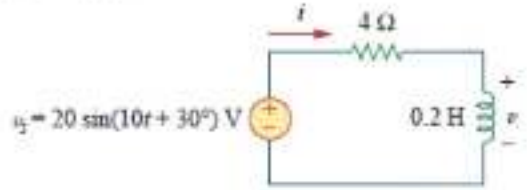
$$Y = G + jB = \frac{1}{Z} = \frac{1}{R + jX} = \frac{R - jX}{(R + jX)(R - jX)} = \frac{R - jX}{R^2 + X^2} = \frac{R}{R^2 + X^2} - j\frac{X}{R^2 + X^2}$$

Or:

$$G = \frac{R}{R^2 + X^2} \text{ and } B = -\frac{X}{R^2 + X^2}$$

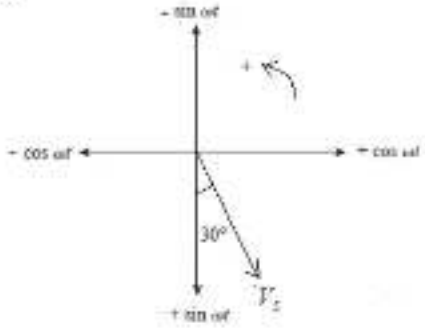
Example:

Find $i(t)$ and $v(t)$ for the circuit shown.



Solution:

From the diagram below, we find:



$v_s(t) = 20\sin(10t + 30^0) = 20\cos(10t - 60^0)$ or $V_s = 20\angle -60^0 V$ and $\omega = 10 rad/s$

$Z = R + j\omega L = 4 + j(10)(0.2) = 4 + j2 = 4.47\angle 26.56^0$

$$I = \frac{V}{Z} = \frac{20\angle -60^0}{4.47\angle 26.56^0} = 4.47\angle -86.56^0 A$$

Or:

$$i(t) = 4.47\cos(10t - 86.56^0)A$$
$$= 4.47\sin(10t + 3.44^0)A$$

$$V = X.I = (j\omega L).I = j(10x0.2)(4.47\angle -86.56^0)$$
$$= (2\angle 90^0)(4.47\angle -86.56^0) = 8.94\angle 3.44^0 V$$

Or:

$$v(t) = 8.94\cos(10t - 3.44^0)V = 8.94\sin(10t + 93.44^0)V$$

Kirchhoff's Laws in the Frequency Domain

Kirchhoff's voltage law (KVL) states that the sum of the voltages in a closed loop is zero, or:

$\sum_{i=1}^{n} v_i(t) = 0$, if

$v_i(t) = V_{mi}\cos(\omega t + \theta_i)$, thus:

$\sum_{i=1}^{n} V_{mi}\cos(\omega t - \theta_i) = 0$, or:

$V_{m1}\cos(\omega t + \theta_1) + V_{m2}\cos(\omega t + \theta_2) + \ldots + V_{mn}\cos(\omega t - \theta_n) = 0$

This can be written as:

$\mathrm{Re}\left\{V_{m1}e^{j(\omega t+\theta_1)} + V_{m2}e^{j(\omega t+\theta_2)} + \ldots + V_{mn}e^{j(\omega t+\theta_n)}\right\} = 0$

$\mathrm{Re}\left\{V_{m1}e^{j\omega t}.e^{j\theta_1} + V_{m2}e^{j\omega t}.e^{j\theta_2} + \ldots + V_{mn}e^{j\omega t}.e^{j\theta_n}\right\} = 0$

$e^{j\omega t}.\mathrm{Re}\left\{V_{m1}e^{j\theta_1} + V_{m2}e^{j\theta_2} + \ldots + V_{mn}e^{j\theta_n}\right\} = 0$

As, $e^{j\omega t} \neq 0$, thus:

$\mathrm{Re}\left\{V_{m1}e^{j\theta_1} + V_{m2}e^{j\theta_2} + \ldots + V_{mn}e^{j\theta_n}\right\} = 0$

This can be written in phasor form:

$V_1 + V_2 + \ldots + V_n = 0$

Or, Kirchhoff's voltage law (KVL) is valid in phasor form.

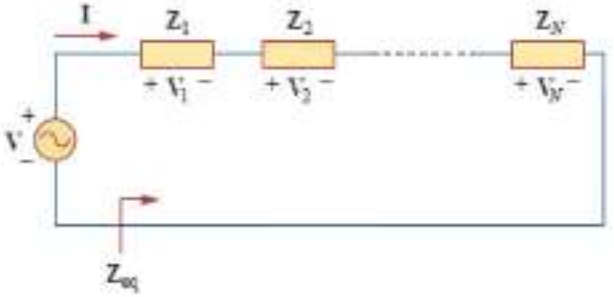
Following a similar procedure, we can show that Kirchhoff's current law (KCL) holds for phasors, or:

$I_1 + I_2 + \ldots + I_n = 0$

Once we have shown that both KVL and KCL hold in the frequency domain, it is easy to do many things, such as impedance combination, nodal and mesh analyses, superposition, and source transformation.

Impedance Combinations

Series impedances: consider a circuit with serially connected impedances, shown below.



Applying KVL for this circuit:

$$V = V_1 + V_2 + ... + V_n$$

Or:

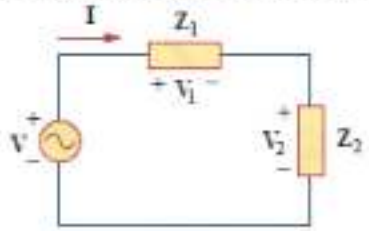
$$V = IZ_1 + IZ_2 + ... + IZ_n = I(Z_1 + Z_2 + ... + Z_n)$$

As, $V = IZ_{eq}$, thus:

$$Z_{eq} = Z_1 + Z_2 + ... + Z_n$$

This indicates that the total or equivalent impedance of series-connected impedances is the sum of the individual impedances.

For the case of two serially connected impedances, shown below.



$$Z_{eq} = Z_1 + Z_2$$

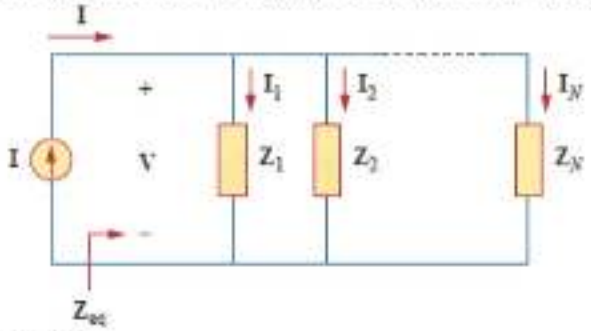
$$I = \frac{V}{Z_{eq}} = \frac{V}{Z_1 + Z_2}$$

And:

$$V_1 = IZ_1 = \frac{VZ_1}{Z_1 + Z_2} = \frac{Z_1}{Z_1 + Z_2}V$$

$$V_2 = IZ_2 = \frac{Z_2}{Z_1 + Z_2}V$$

Parallel impedances: consider a circuit with parallel connected impedances, shown below.



Applying KCL for this circuit:

$$I = I_1 + I_2 + ... + I_n$$

Or:

$$I = \frac{V}{Z_1} + \frac{V}{Z_2} + ... + \frac{V}{Z_n} = V\left(\frac{1}{Z_1} + \frac{1}{Z_2} + ... + \frac{1}{Z_n}\right)$$

As, $I = \frac{V}{Z_{eq}} = V.\frac{1}{Z_{eq}}$, thus:

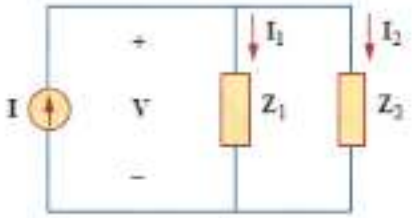
$$\frac{1}{Z_{eq}} = \frac{1}{Z_1} + \frac{1}{Z_2} + ... + \frac{1}{Z_n}$$

and the equivalent admittance is:

$$Y_{eq} = Y_1 + Y_2 + ... + Y_n$$

This indicates that the equivalent admittance of parallel-connected admittances is the sum of the individual admittances.

For the case of two parallel connected impedances, shown below.



$$\frac{1}{Z_{eq}} = \frac{1}{Z_1} + \frac{1}{Z_2} = \frac{Z_1 + Z_2}{Z_1.Z_2}$$

Or:

$$Z_{eq} = \frac{Z_1.Z_2}{Z_1 - Z_2}$$

And:

$$V = I.Z_{eq} = I\frac{Z_1.Z_2}{Z_1 + Z_2} = I_1Z_1 = I_2Z_2$$

From this we deduce:

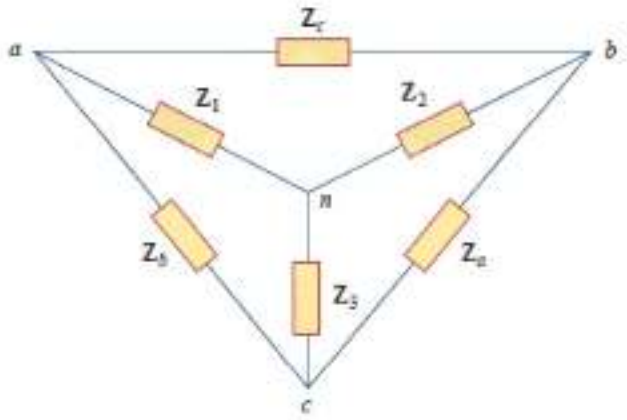
$$I\frac{Z_1.Z_2}{Z_1 + Z_2} = I_1Z_1$$

Or:

$I_1 = I\frac{Z_2}{Z_1 + Z_2}$ and $I_2 = I\frac{Z_1}{Z_1 - Z_2}$, which is the current-division principle.

Delta-to-wye and wye-to-delta transformations

Consider the following circuit with impedances Z_1, Z_2 and Z_3 connected in wye (star) connection and their equivalent impedances Z_a, Z_b and Z_c connected in delta connection.



From this circuit:

$$Z_{ab} = Z_C \,//\, (Z_a + Z_b) = \frac{Z_C.(Z_a + Z_b)}{Z_a + Z_b + Z_C}$$

Also:

$$Z_{ab} = Z_1 + Z_2$$

Or:

$$Z_1 + Z_2 = \frac{Z_C.(Z_a + Z_b)}{Z_a + Z_b + Z_C} \qquad (1)$$

In the same way, we find:

$$Z_2 + Z_3 = \frac{Z_a.(Z_b + Z_C)}{Z_a + Z_b + Z_C} \qquad (2)$$

$$Z_1 + Z_3 = \frac{Z_b.(Z_a + Z_C)}{Z_a + Z_b + Z_C} \qquad (3)$$

Subtracting (2) from (1):

$$Z_1 - Z_3 = \frac{Z_C.(Z_a + Z_b)}{Z_a + Z_b + Z_C} - \frac{Z_a.(Z_b + Z_C)}{Z_a + Z_b + Z_C} = \frac{Z_C.(Z_a + Z_b) - Z_a.(Z_b + Z_C)}{Z_a + Z_b + Z_C} \qquad (4)$$

Adding (3) to (4), we find:

$$2Z_1 = \frac{Z_b.(Z_a + Z_C)}{Z_a + Z_b + Z_C} + \frac{Z_C.(Z_a + Z_b) - Z_a.(Z_b + Z_C)}{Z_a + Z_b + Z_C}$$

$$2Z_1 = \frac{Z_b.(Z_a + Z_C) + Z_C.(Z_a + Z_b) - Z_a.(Z_b + Z_C)}{Z_a + Z_b + Z_C}$$

$$2Z_1 = \frac{Z_a Z_b - Z_b Z_C + Z_a Z_C + Z_b Z_C - Z_a Z_b - Z_a Z_C}{Z_a + Z_b + Z_C}$$

$$2Z_1 = \frac{2Z_b Z_C}{Z_a + Z_b + Z_C}$$

$$Z_1 = \frac{Z_b Z_C}{Z_a + Z_b + Z_C}$$

In the same way, we can find that:

$$Z_2 = \frac{Z_a Z_c}{Z_a + Z_b + Z_c}$$

$$Z_3 = \frac{Z_a Z_b}{Z_a + Z_b + Z_c}$$

These equations can be used to find the equivalent star-connection for delta-connected impedances.

For $Y - \Delta$ conversion, we can find that:

$$Z_a = \frac{Z_1 Z_2 + Z_2 Z_3 + Z_1 Z_3}{Z_1}$$

$$Z_b = \frac{Z_1 Z_2 + Z_2 Z_3 + Z_1 Z_3}{Z_2}$$

$$Z_c = \frac{Z_1 Z_2 + Z_2 Z_3 + Z_1 Z_3}{Z_3}$$

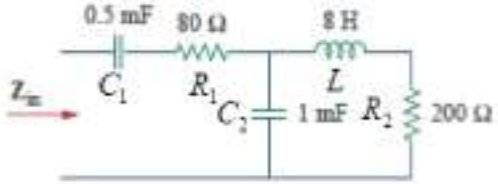
For balanced circuits (circuits with equal impedances), we find that:

$$Z_Y = \frac{Z_\Delta Z_\Delta}{Z_\Delta + Z_\Delta + Z_\Delta} = \frac{Z_\Delta}{3}$$

And:

$$Z_\Delta = 3Z_Y$$

Example: Find the input impedance of the circuit in the figure below at $\omega = 10 rad/s$



Solution:

$$Z_1 = R_1 + \frac{1}{j\omega C_1} = 80 + \frac{1}{j(10)(0.5x10^{-3})} = 80 - j200$$

$$Z_2 = \frac{1}{j\omega C_2} = \frac{1}{j(10)(1x10^{-3})} = -j100$$

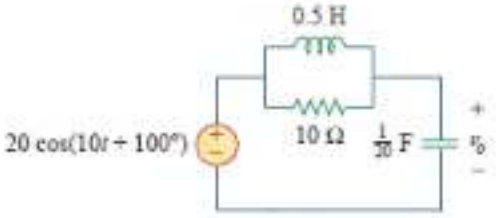
$$Z_3 = R_2 + j\omega L = 200 + j(10)(8) = 200 + j80$$

$$Z_{in} = Z_1 + Z_2 // Z_3 = Z_1 + \frac{Z_2 Z_3}{Z_2 + Z_3}$$

$$Z_{in} = 80 - j200 + \frac{-j100(200 + j80)}{-j100 + 200 + j80}$$

$$Z_{in} = 80 - j200 + \frac{8000 - j20000}{200 - j20} = 129.52 - j295$$

Example: Find v_0 in the circuit shown.



$$Z_1 = \frac{R(j\omega L)}{R - j\omega L} = \frac{10(j10x0.5)}{10 + j(10x0.5)} = \frac{j50(10 - j5)}{(10 + j5)(10 - j5)} = 2 + j4$$

$$Z_2 = \frac{1}{j\omega C} = \frac{1}{j(10x0.05)} = -j2$$

$$V_0 = \frac{Z_2 V}{Z_1 - Z_2} = \frac{-j2(20\angle 100^\circ)}{2 + j4 - j2} = \frac{(2\angle -90^\circ)(20\angle 100^\circ)}{2 + j2} = \frac{40\angle 10^\circ}{2\sqrt{2}\angle 45^\circ} = 14.142\angle -35^\circ$$

Or:

$$v_0(t) = 14.142\cos(10t - 35^\circ)$$

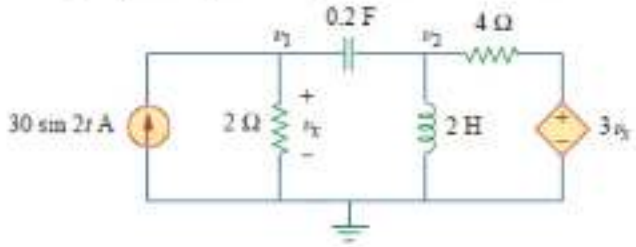
Sinusoidal Steady-State Analysis

Methods used for DC analysis can be used for AC analysis using phasor representation for voltage and current values. In other words, nodal analysis, mesh analysis, Thevenin's theorem, Norton's theorem, superposition, and source transformations are applied in analyzing ac circuits.

Nodal Analysis

This method is based on Kirchhoff's current law (KCL).

Example 1: Using nodal analysis, find v_1 and v_2 in the circuit below.



Solution:

The current source: $I = 30\angle -90^{\circ} = -30j$, $\omega = 2$

The capacitor: $X_C = \dfrac{1}{j\omega C} = -j\dfrac{1}{(2)(0.2)} = -2.5j$

The inductor: $X_L = j\omega L = j(2)(2) = 4j$

For node 1:

$$\frac{V_1}{2} + \frac{V_1 - V_2}{-2.5j} = -30j$$

Or:

$$(2 - 2.5j)V_1 - 2V_2 = -150 \qquad (1)$$

For node 2:

$$\frac{V_2 - V_1}{-2.5j} + \frac{V_2}{4j} + \frac{V_2 - 3V_x}{4} = 0$$

but $V_x = V_1$, thus:

$$\frac{V_2 - V_1}{-2.5j} + \frac{V_2}{4j} + \frac{V_2 - 3V_1}{4} = 0$$

Or:

$$(-4 + 7.5j)V_1 + (1.5 - 2.5j)V_2 = 0 \qquad (2)$$

Solving (1) and (2), using Cramer's rule:

$$\Delta = \begin{vmatrix} 2 - 2.5j & -2 \\ -4 + 7.5j & 1.5 - 2.5j \end{vmatrix} = -11.25 + 6.25j$$

$$\Delta_1 = \begin{vmatrix} -150 & -2 \\ 0 & 1.5 - 2.5j \end{vmatrix} = -225 - 375j$$

And:

$$V_1 = \frac{\Delta_1}{\Delta} = 29.434 - 16.981j = 33.98\angle -29.98^\circ$$

Or:

$$v_1(t) = 33.98\cos(2t - 29.98^\circ) = 33.98\sin(2t + 60.02^\circ)$$

And:

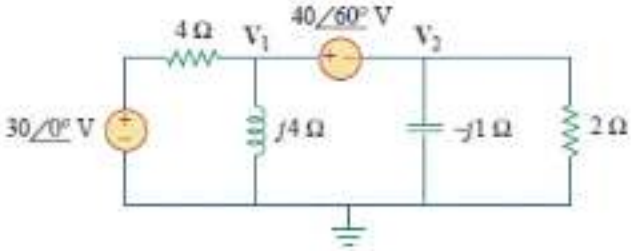
$$\Delta_2 = \begin{vmatrix} 2 - 2.5j & -150 \\ -4 + 7.5j & 0 \end{vmatrix} = -600 + 1125j$$

$$V_2 = \frac{\Delta_2}{\Delta} = -83.208 - 53.774j = 99.071\angle -32.87^\circ$$

Or:

$$v_2(t) = 99.071\cos(2t - 32.87^\circ) = 99.071\sin(2t + 57.13^\circ)$$

Example 2: Using nodal analysis, find v_1 and v_2 in the circuit below.



Solution:

Nodes 1 and 2 form a supernode. Applying KCL at the supernode gives:

$$\frac{V_1 - 30\angle 0^\circ}{4} + \frac{V_1}{4j} + \frac{V_2}{-1j} + \frac{V_2}{2} = 0$$

or:

$$(1 + j)V_1 + (-4 + 2j)V_2 = 30j \qquad (1)$$

but:

$$V_1 - V_2 = 40\angle 60^\circ = 20 + j20\sqrt{3} \qquad (2)$$

Solving (1) and (2), using Cramer's rule:

$$\Delta = \begin{vmatrix} 1 + j & -4 + 2j \\ 1 & -1 \end{vmatrix} = 3 - 3j$$

$$\Delta_1 = \begin{vmatrix} 30j & -4 + 2j \\ 20 + j20\sqrt{3} & -1 \end{vmatrix} = 149.28 + 68.56j$$

And:

$$V_1 = \frac{\Delta_1}{\Delta} = 13.45 + 36.31j = 38.72\angle 69.67^\circ$$

And:

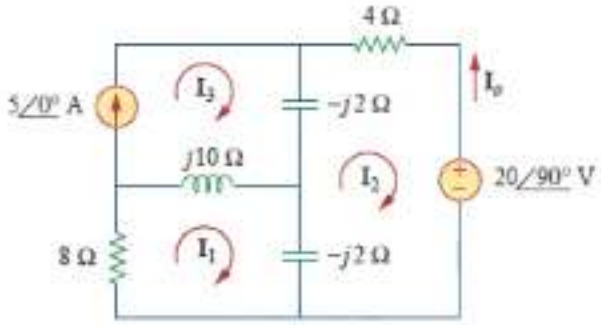
$$\Delta_2 = \begin{vmatrix} 1+j & 30j \\ 1 & 20+j20\sqrt{3} \end{vmatrix} = -14.64 + 24.64j$$

$$V_2 = \frac{\Delta_2}{\Delta} = -6.55 - 1.67j = 6.76\angle 165.7^\circ$$

Mesh Analysis

This method is based on Kirchhoff's voltage law (KVL).

Example 1: Using mesh analysis, find I_o in the circuit below.



Solution:

For mesh 1:

$$8I_1 + j10(I_1 - I_3) - j2(I_1 - I_2) = 0$$

Or:

$$(8 + j8)I_1 + j2I_2 - j10I_3 = 0 \qquad (1)$$

For mesh 2:

$$-j2(I_2 - I_1) - j2(I_2 - I_3) + 4I_2 = -20\angle 90^\circ$$

Or:

$$j2I_1 + (4 - j4)I_2 + j2I_3 = -j20 \qquad (2)$$

Also:

$$I_3 = 5 \qquad (3)$$

Substituting (3) in (1) and (2), we find:

$$(8 + j8)I_1 + j2I_2 = j50 \qquad (4)$$

$$j2I_1 + (4 - j4)I_2 = -j30 \qquad (5)$$

Solving (4) and (5) using Cramer's rule, we find:

$$\Delta = \begin{vmatrix} 8+j8 & j2 \\ j2 & 4-j4 \end{vmatrix} = 68$$

$$\Delta_1 = \begin{vmatrix} j50 & j2 \\ -j30 & 4-j4 \end{vmatrix} = 140 + j200$$

$$I_1 = \frac{\Delta_1}{\Delta} = 2.06 + j2.94$$

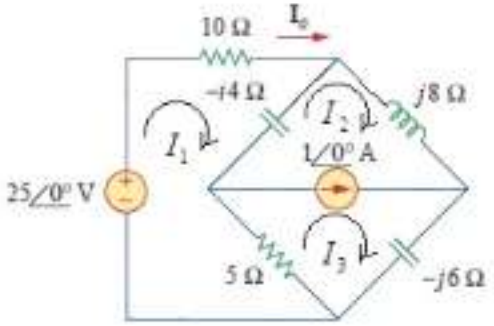
$$\Delta_2 = \begin{vmatrix} 8+j8 & j50 \\ j2 & -j30 \end{vmatrix} = 340 - j240$$

$$I_2 = \frac{\Delta_2}{\Delta} = 5 - j3.53$$

And:

$$I_0 = -I_2 = -5 + j3.53 = 6.12\angle 144.78^\circ\ A$$

Example 1: Using mesh analysis, find I_o in the circuit below.



For mesh 1:

$$10I_1 - j4(I_1 - I_2) + 5(I_1 - I_3) = 25$$

Or:

$$(15 - j4)I_1 + j4I_2 - 5I_3 = 25 \qquad (1)$$

Meshes 2 and 3 form a super-mesh, for this super mesh:

$$j8I_2 - j6I_3 + 5(I_3 - I_1) - j4(I_2 - I_1) = 0$$

Or:

$$(-5 + j4)I_1 + j4I_2 + (5 - j6)I_3 = 0 \qquad (2)$$

For the super-mesh:

$$I_2 - I_3 = -1 \qquad (3)$$

Equations (1), (2) and (3) can be solved by substitution or by Gaussian elimination or by Cramer's rule:

Using Cramer's rule:

$$\Delta = \begin{vmatrix} 15 - j4 & j4 & -5 \\ -5 + j4 & j4 & 5 - j6 \\ 0 & 1 & -1 \end{vmatrix} = -58 + j10$$

$$\Delta_1 = \begin{vmatrix} 25 & j4 & -5 \\ 0 & j4 & 5 - j6 \\ -1 & 1 & -1 \end{vmatrix} = -149 + j10$$

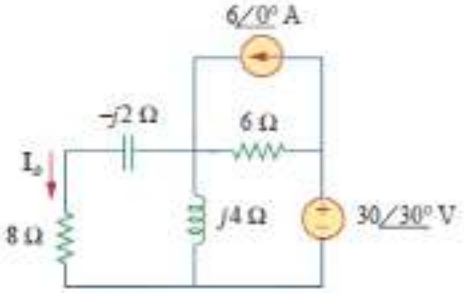
And:

$$I_0 = I_1 = \frac{\Delta_1}{\Delta} = 2.52 + j0.263 = 2.537\angle 5.93^0\ A$$

Superposition Theorem

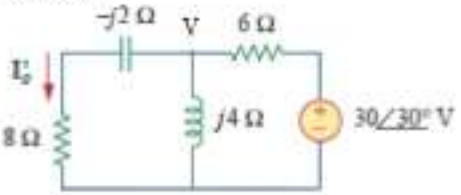
The superposition principle is applicable because electrical circuits are linear. It is important for analyzing circuits having sources with different frequencies. The total response is obtained by adding the individual responses in the time domain. It is incorrect to try to add the responses in the phasor or frequency domain, because phasors are rotating at different angular frequencies. Thus, when a circuit has sources operating at different frequencies, one must add the responses due to the individual frequencies in the time domain.

Example: Find current I_0 in the circuit below, using the superposition theorem.



Solution:

- The component of I_0 caused by the voltage source is found from the following circuit, in which the current source is removed.



Using nodal analysis:

$$\frac{V-30\angle 30^{o}}{6}+\frac{V}{j4}+\frac{V}{8-j2}=0$$

Or:

$$\frac{V-30\left(\cos 30^{o}+j\sin 30^{o}\right)}{6}+\frac{V}{j4}+\frac{V}{8-j2}=0$$

$$\frac{V-30\left(\frac{\sqrt{3}}{2}+j0.5\right)}{6}+\frac{V}{j4}+\frac{V}{8-j2}=0$$

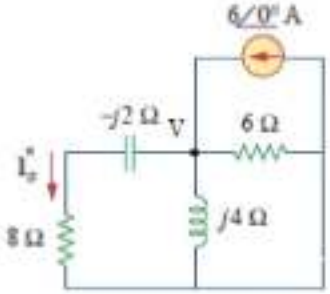
From which:

$V = 5.25 + j12.87$

And:

$$I_0=\frac{V}{8-j2}=\frac{5.25+j12.87}{8-j2}=0.24+j1.67\ \ A$$

- The component of I_0 caused by the current source is found from the following circuit, in which the voltage source is short-circuited.



Using again nodal analysis:

$$\frac{V}{8-j2}+\frac{V}{j4}+\frac{V}{6}-6$$

From which:

$$V=13.17+10.22 \quad V$$

And:

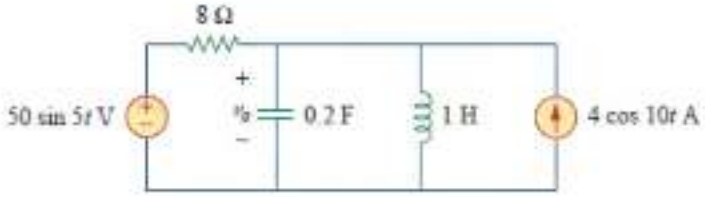
$$I_0^{'}=\frac{V}{8-j2}=\frac{13.17+10.22}{8-j2}=1.25-j1.59 \quad A$$

And:

$$I=I_0+I_0^{'}=1.49+j3.26 \quad A=3.58\angle 65.44 \quad A$$

Example:

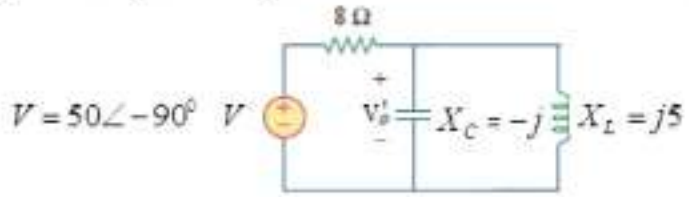
Calculate v_0 in the following circuit, using the superposition theorem.



Solution:

This circuit have two sources with different frequencies.

- the component of v_0 caused by the voltage source is found from the following circuit.



For this circuit:

$V = 50\angle -90^{0}\ \ V, \qquad \omega = 5rad/s\ ,\ \ X_L = j\omega L = j5\ \ \Omega, \quad X_C = \dfrac{1}{j\omega C} = -j\ \ \Omega$

$$V_0' = \frac{V}{R - X_C /\!/ X_L}(X_C /\!/ X_L)$$

Where:

$$X_C /\!/ X_L = \frac{X_C . X_L}{X_C + X_L} = \frac{-j(j5)}{j4} = -j1.25\ \ \Omega$$

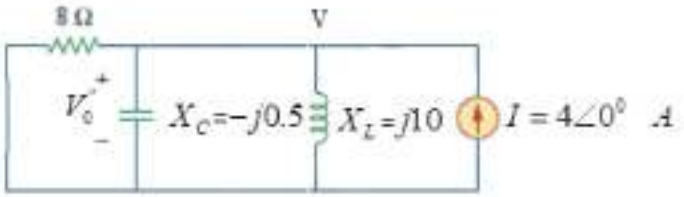
Thus:

$$V_0' = \frac{-j50}{8 - j1.25}(-j1.25) = -7.63 - j1.19 = 7.72\angle 188.88^{0}\ \ V \text{ and } \omega = 5rad/s$$

Or:

$$v_0(t) = 7.72\cos(5t + 188.88^{0}) = 7.72\sin(5t - 81.12^{0})\ \ V$$

- the component of v_0 caused by the current source is found from the following circuit.



For this circuit:

$$I = 4\angle 0^{0}\ \ A, \qquad \omega = 10rad/s\ ,\ \ X_L = j\omega L = j10\ \ \Omega, \quad X_C = \frac{1}{j\omega C} = -j0.5\ \ \Omega$$

Using nodal analysis:

$$\frac{V}{X_L} + \frac{V}{X_C} + \frac{V}{R} = I$$

Or:

$$\frac{V}{j10} - \frac{V}{j0.5} + \frac{V}{8} = 4$$

From which:

$$V_0' = V = 0.14 - j2.1 = 2.1\angle -86.24^{0}\ \ V \text{ and } \omega = 10rad/s$$

Or:

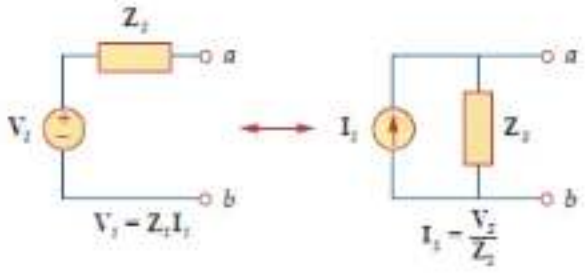
$$v_0'(t) = 2.1\cos(10t - 86.24^{0})\ \ V$$

Thus:

$$v_0(t) = v_0'(t) + v_0''(t) = 7.72\sin(5t - 81.12^{0}) + 2.1\cos(10t - 86.24^{0})$$

Source Transformation

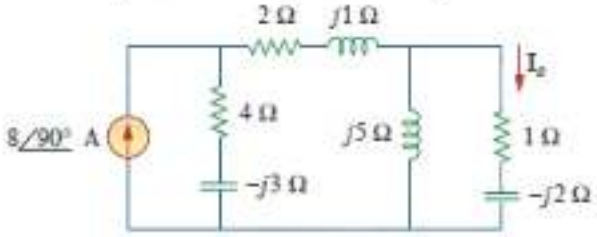
Here, a voltage source in series with an impedance is transformed into a current source in parallel with an impedance, or vice versa, as seen below.



Here:

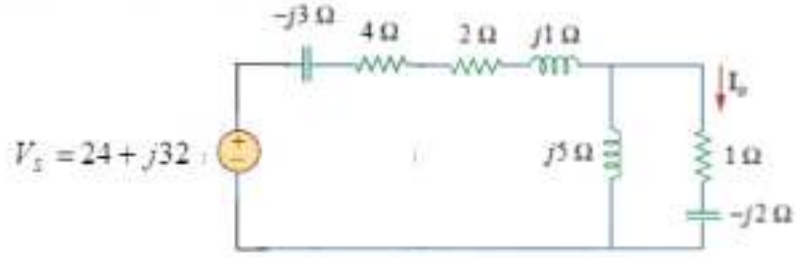
$$I_S = \frac{V_S}{Z_S} \qquad \text{or} \qquad V_S = Z_S . V_S$$

Example: Find I_o in the following circuit, using the concept of source transformation.



Solution:

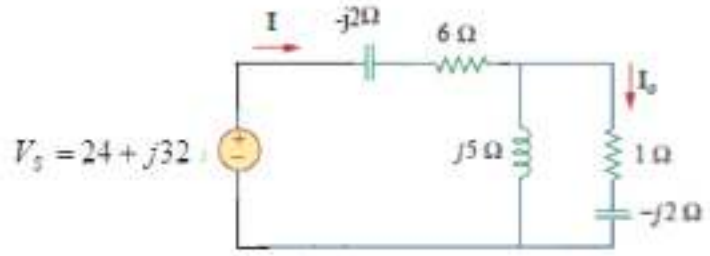
The current source with the parallel impedance $(4 - j3)\Omega$ is transformed into a voltage source with a series impedance as shown below.



Here:

$V_s = I_s . Z_s = j8(4 - j3) = 24 + j32 \;\; V$

This circuit is reduced to:



The impedance seen by the voltage source is:

$$Z = (6 - j2) + \frac{j5(1 - j2)}{j5 + (1 - j2)} = 8.5 - j4.5$$

And the current I is:

$$I = \frac{V_s}{Z} = \frac{24 - j32}{8.5 - j4.5} = 0.65 + j4.11 \quad A$$

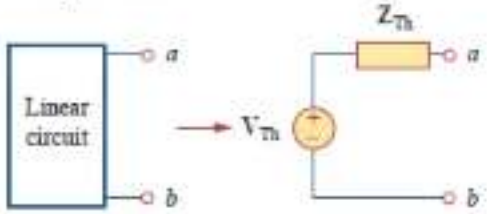
The current I_o is found using the current division rule, or:

$$I_o = I\frac{(j5)}{j5 + (1 - j2)} = \frac{(0.65 + j4.11)(j5)}{1 + j3} = -1.08 + j6.49 = 6.576\angle 99.46^\circ \quad A$$

Thevenin and Norton Equivalent Circuits

Thevenin equivalent circuit

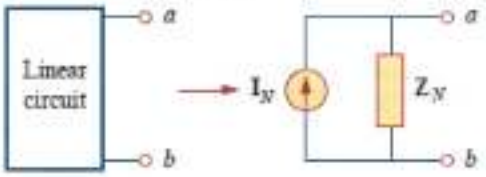
Thevenin equivalent circuit is depicted in the figure below, where a linear circuit is replaced by a voltage source in series with an impedance.



Where V_{Th} is the open-circuit voltage.

Norton equivalent circuit

The Norton equivalent circuit is illustrated in the figure below, where a linear circuit is replaced by a current source in parallel with an impedance.



Where I_N is the short-circuit current.

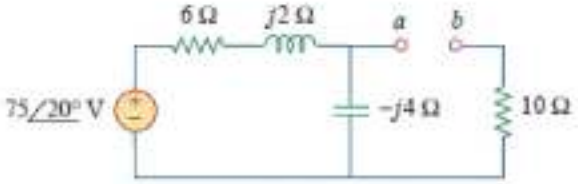
Keep in mind that the two equivalent circuits are related as

$Z_{Th} = Z_N$ and $V_{Th} = Z_N I_N$

Note that, if the circuit has sources operating at different frequencies, the Thevenin or Norton equivalent circuits must be determined at each frequency. This leads to entirely different equivalent circuits, one for each frequency.

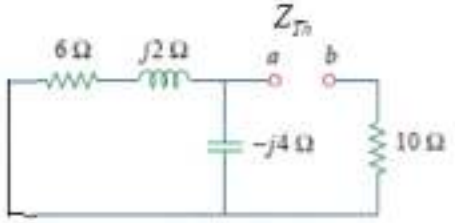
Example 1:

Find the Thevenin equivalent at terminals a-b of the circuit in the figure below.



Solution:

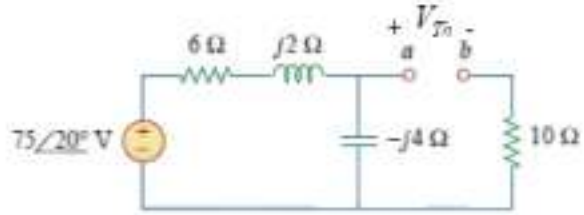
The impedance Z_{Th} is found after removing the voltage source, as in the following circuit.



From this circuit:

$$Z_{Th} = \frac{(6+j2)(-j4)}{6+j2-j4} + 10 = 12.4 - j3.2 \ \ \Omega$$

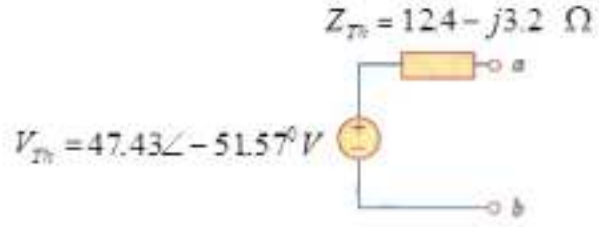
The voltage V_{Th} is found from the following circuit.



It can be seen that V_{Th} is equal to the voltage on the capacitor, or:

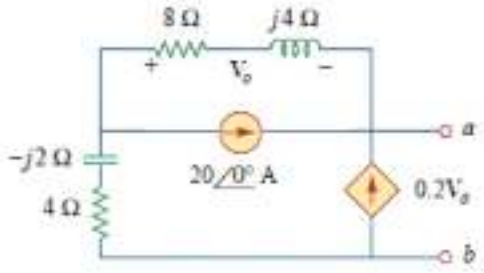
$$V_{Th} = \frac{75\angle 20^\circ}{6+j2-j4}(-j4) = \frac{70.48+j25.65}{6+j2-j4}(-j4) = 29.49 - j37.16 = 47.43\angle -51.57^\circ V$$

Or the Thevenin equivalent circuit is:



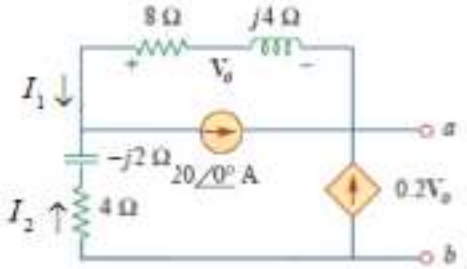
Example 2:

Determine the Thevenin equivalent of the circuit shown below, as seen from the terminals *a-b*.



Solution:

The voltage V_{Th} is found from the following circuit.



For this circuit:

$$I_1 + I_2 = 20 \qquad (1)$$

$$V_o = -I_1(8 + j4) \qquad (2)$$

And:

$$I_2 = -0.2V_o \qquad (3)$$

From (2) and (3);

$$I_2 = 0.2I_1(8 + j4) \qquad (4)$$

Substituting (4) in (1):

$$I_1 + 0.2I_1(8 + j4) = 20$$

Or:

$$(2.6 + j0.8)I_1 = 20$$

Or:

$$I_1 = \frac{20}{2.6 - j0.8} = 7.03 - j2.16$$

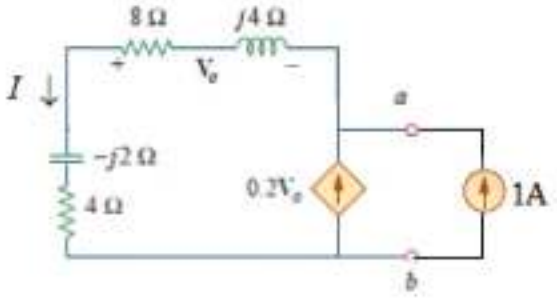
And:

$$I_2 = 20 - I_1 = 12.97 + j2.16$$

And:

$$V_{Th} = I_1(8 + j4) - I_2(4 - j2) = 8.65 - j28.11 = 29.41\angle 72.9^\circ V$$

To find Z_{Th} we remove the independent current source and assume a current source of 1A at the circuit terminals as shown below.



From this circuit:

$$V_{ab} = I(8 + j4 + 4 - j2) = I(12 + j2) \qquad (1)$$

And:

$$I = 1 + 0.2V_o = 1 - 0.2I(8 + j4) \qquad (2)$$

From which:

$$I + 0.2I(8 + j4) = 1$$

$$(2.6 + j0.8)I = 1$$

Or:

$$I = \frac{1}{2.6 + j0.8} = 0.3514 - j0.108 \qquad (3)$$

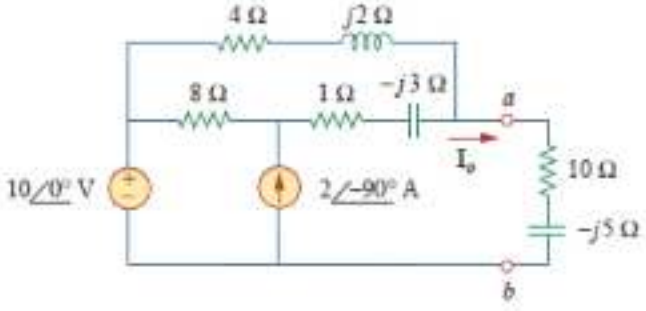
Substituting (3) in (1), we find:

$$V_{ab} = (0.3514 - j0.108)(12 + j2) = 4.43 - j0.59 = 4.473\angle -7.64^\circ V$$

And:

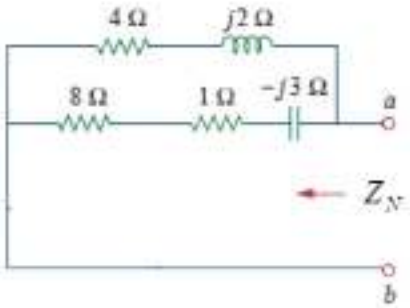
$$Z_{Th} = \frac{V_{ab}}{1} = 4.473\angle -7.64^\circ \Omega$$

Example 3: Determine the Norton equivalent of the circuit shown below, as seen from terminals a-b. Use the equivalent to find Io.



Solution:

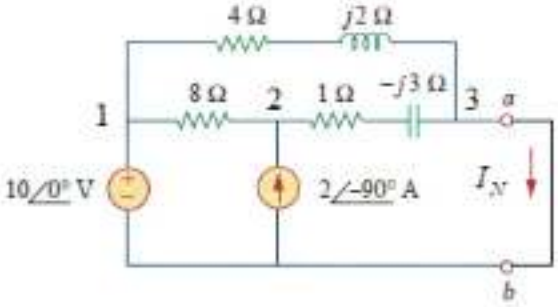
Z_N is found after removing the sources, as below.



From this circuit:

$$Z_N = \frac{(9-j3)(4+j2)}{9-j3+4+j2} = 3.176 + j0.706 \ \ \Omega$$

I_N is founded by short-circuiting the terminals a-b, as below.



For this circuit:

$V_1 = 10V$, $V_3 = 0V$ and by nodal analysis:

$$\frac{V_2 - V_1}{8} + \frac{V_2 - V_3}{1-j3} = -j2$$

Or:

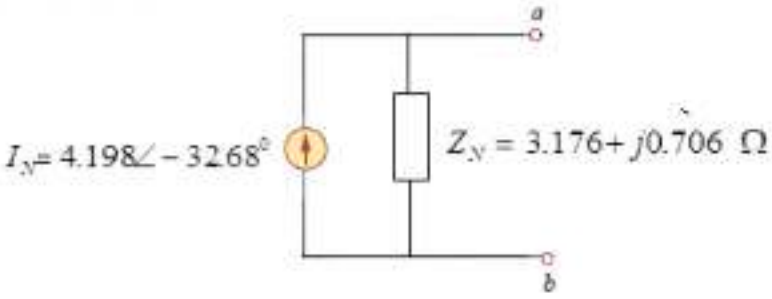
$$\frac{V_2 - V_1}{8} + \frac{V_2 - V_3}{1-j3} = -j2$$

From which:

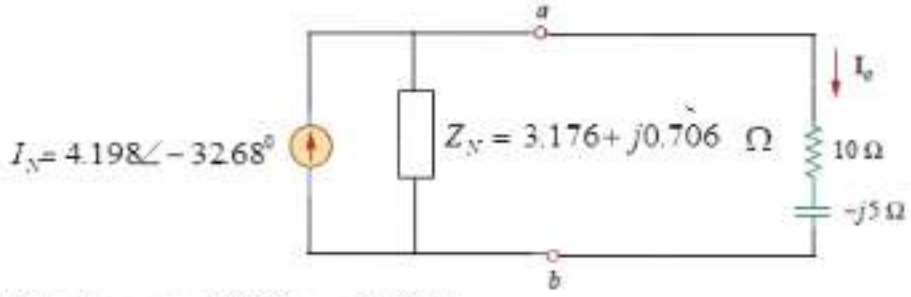
$$V_2 = -2.267 - j5.87 \ \ V$$

$$I_N = \frac{V_2 - V_3}{1-j3} + \frac{V_1 - V_3}{4+j2} = 3.533 - j2.267 = 4.198\angle -32.68^\circ$$

Or Norton equivalent circuit is:



The current I_o is found using Norton equivalent, as below.



Using the current division principle:

$$I_o = I_N \frac{Z_N}{Z_N + (10 - j5)} = 0.985 - j0.036 = 0.986\angle -2.1^0 \quad A$$

AC Power Analysis

Instantaneous and Average Power

The instantaneous power $p(t)$ absorbed by an element is the product of the instantaneous voltage $v(t)$ across the element and the instantaneous current $i(t)$ through it, or:

$p(t) = v(t).i(t)$

It is the power at any instant of time.

Let the voltage and current at the terminals of the circuit be

$v(t) = V_m \cos(\omega t + \theta_v)$

$i(t) = I_m \cos(\omega t + \theta_i)$

Thus:

$$p(t) = v(t).i(t) = V_m \cos(\omega t + \theta_v).I_m \cos(\omega t + \theta_i)$$

$$= V_m.I_m \cos(\omega t + \theta_v)\cos(\omega t + \theta_i)$$

$$= \frac{V_m.I_m}{2}\left[\cos(\theta_v - \theta_i) + \cos(2\omega t + \theta_v + \theta_i)\right] \qquad (1)$$

$$= \frac{V_m.I_m}{2}\left[\cos(\theta_v - \theta_i)\right] - \frac{V_m.I_m}{2}\left[\cos(2\omega t + \theta_v + \theta_i)\right]$$

This shows us that the instantaneous power has two parts. The first part is constant or time independent. Its value depends on the phase difference between the voltage and the current. The second part is a sinusoidal function whose frequency is twice the angular frequency of the voltage or current

For example, a sketch of $p(t)$ for $v(t) = V_m \sin(\omega t)$ and $i(t) = I_m \sin(\omega t - \theta_i)$ is shown below



We observe that $p(t)$ is periodic and has a frequency that is twice that of voltage or current. We also observe that $p(t)$ is positive for some part of each cycle and negative for the rest of the cycle. When $p(t)$ is positive, power is absorbed by the circuit. When $p(t)$ is negative, power is absorbed by the source; that is, power is transferred from the circuit to the source. This is possible because of the storage elements (capacitors and inductors) in the circuit.

The average power over one period is:

$$P = \frac{1}{T}\int_0^T p(t).dt \qquad (2)$$

Substituting (1) in (2), we have:

$$P = \frac{1}{T}\int_0^T \left\{ \frac{V_m.I_m}{2}\left[\cos(\theta_v - \theta_i)\right] + \frac{V_m.I_m}{2}\left[\cos(2\omega t + \theta_v + \theta_i)\right] \right\} dt$$

$$P = \frac{1}{T}\int_0^T \frac{V_m.I_m}{2}\left[\cos(\theta_v - \theta_i)\right]dt + \frac{1}{T}\int_0^T \frac{V_m.I_m}{2}\left[\cos(2\omega t + \theta_v + \theta_i)\right]dt$$

$$P = \frac{V_m.I_m}{2}\cos(\theta_v - \theta_i)\frac{1}{T}\int_0^T dt + \frac{V_m.I_m}{2}\frac{1}{T}\int_0^T \cos(2\omega t + \theta_v + \theta_i)dt$$

The second integrand vanishes as the average of a sinusoid over its period is zero because the area under the sinusoid during a positive half-cycle is canceled by the area under it during the following negative half-cycle. Thus the average power becomes:

$$P = \frac{V_m.I_m}{2}\cos(\theta_v - \theta_i)\frac{1}{T}\int_0^T dt = \frac{V_m.I_m}{2}\cos(\theta_v - \theta_i) \qquad (3)$$

If $v(t)$ and $i(t)$ are represented in phasor forms, or, if:

$V = V_m \angle \theta_v$ and $I = I_m \angle \theta_i$, thus:

$$\begin{aligned} V.I^* &= V_m \angle \theta_v . I_m \angle -\theta_i = V_m.I_m \angle(\theta_v - \theta_i) \\ &= V_m.I_m\left[\cos(\theta_v - \theta_i) + j\sin(\theta_v - \theta_i)\right] \end{aligned} \qquad (4)$$

Or:

$$\mathrm{Re}\{V.I^*\} = V_m.I_m.\cos(\theta_v - \theta_i) \qquad (5)$$

Comparing (3) and (5), we conclude that:

$$P = \frac{1}{2}V_m.I_m.\cos(\theta_v - \theta_i) = \frac{1}{2}\mathrm{Re}\{V.I^*\} \qquad (6)$$

Consider two special cases:

- Pure resistive element, here the voltage and the current are in phase, or $\theta_v = \theta_i$, thus:

$$P = \frac{V_m.I_m}{2}\cos(0) = \frac{V_m.I_m}{2} = \frac{1}{2}V_m.\frac{V_m}{R} = \frac{1}{2}\frac{(V_m)^2}{R}$$

Also:

$$P = \frac{V_m.I_m}{2} = \frac{1}{2}I_m.R.I_m = \frac{1}{2}R(I_m)^2$$

- Pure rereactive element, here the voltage and the current are out of phase by 90°, or $\theta_v - \theta_i = \pm 90^\circ$, thus:

$$P = \frac{V_m.I_m}{2}\cos(90^\circ) = 0$$

or a purely reactive circuit absorbs no average power.

Example 1:

If the voltage across the terminals of an element is $v(t) = 165\cos(10t + 20^\circ)\ V$ and the current through it is $i(t) = 20\sin(10t + 60^\circ)\ A$, Calculate the instantaneous power and average power absorbed by this element.

Solution:

$v(t) = 165\cos(10t + 20^\circ)\ V$

$i(t) = 20\sin(10t + 60^\circ) = 20\cos(10t - 30^\circ)$

Thus:

$$p(t) = 165\cos(10t + 20^\circ)\,20\cos(10t - 30^\circ) = 3300\cos(10t + 20^\circ)\cos(10t - 30^\circ)$$

$$= \frac{3300}{2}\left[\cos(50^\circ) + \cos(20t - 10^\circ)\right] = 1061 + 1650\cos(20t - 10^\circ)$$

The active power is $P = 1061W = 1.061kW$.

Example 2:

A current $I = 20\angle 30^\circ\ A$ flows through an impedance $Z = 40\angle -22^\circ\ \Omega$. Find the average power delivered to the impedance.

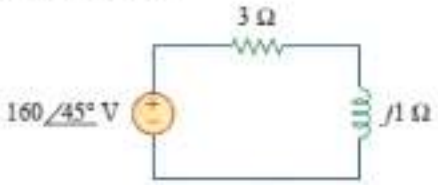
Solution:

$$V = ZI = (40\angle -22^\circ)(20\angle 30^\circ) = 800\angle 8^\circ V$$

$$P = \frac{V_m I_m}{2}\cos(\theta_v - \theta_i) = \frac{(800)(20)}{2}\cos(22^\circ) = 7.417kW$$

Example 3:

In the circuit below, calculate the average power absorbed by the resistor and inductor. Find the average power supplied by the voltage source.



Solution:

$$I = \frac{V_s}{Z} = \frac{160\angle 45^\circ}{3 + 1j} = \frac{160\angle 45^\circ}{\sqrt{10}\angle 18.43^\circ} = 50.6\angle 26.57^\circ A$$

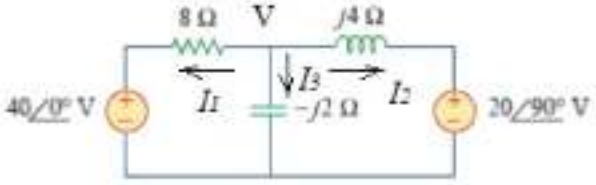
$$P_R = \frac{1}{2}R(I_m)^2 = \frac{1}{2}(3)(50.6)^2 = 3.84kW$$

$$P_L = 0kW$$

$$P_s = \frac{1}{2}V_m I_m\cos(\theta_v - \theta_i) = \frac{1}{2}(160)(50.6)\cos(45^\circ - 26.57^\circ) = 3.84kW$$

Example 4:

Calculate the average power absorbed by each of the five elements in the circuit below.



Solution:

Using nodal analysis:

$$\frac{V-40}{8}+\frac{V}{-j2}+\frac{V-j20}{j4}=0$$

From which $V=16-j32\ \ V$

$$I_1=\frac{V-40}{8}=-3-j4=5\angle 233.13^{0}\ A$$

$$I_2=\frac{V-j20}{j4}=-13-j4=13.6\angle 197.1^{0}\ A$$

$$I_3=\frac{V}{-j2}=16+j8=17.9\angle 26.56^{0}\ A$$

The average power of the resistor is:

$$P_R=\frac{1}{2}R(I_1)^2=\frac{1}{2}8(5)^2=100W$$

The average power of each of the inductor and the capacitor is zero.

The average power generated by the 40V source is:

$$P_{S1}=-\frac{1}{2}V_m.I_1.\cos(\theta_v-\theta_{I1})=-\frac{1}{2}(40)(5)\cos(0-233.13^{0})=60W$$

The average power generated by the 20V source is:

$$P_{S2}=-\frac{1}{2}V_m.I_2.\cos(\theta_v-\theta_{I2})=-\frac{1}{2}(20)(13.6)\cos(90^{0}-197.1^{0})=40W$$

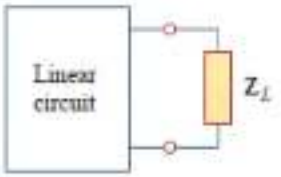
Note that:

$$P_{S1}+P_{S2}=P_R$$

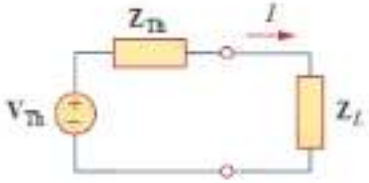
Maximum Average Power Transfer

For DC circuits. it was proved that the maximum power would be delivered to the load if the load resistance is equal to the Thevenin resistance, or if $R_L = R_{Th}$.

Consider an AC linear circuit with Z_L, as shown below.



If this AC linear circuit is represented by its Thevenin's equivalent, the equivalent circuit becomes as below.



For this circuit:

$$I = \frac{V_{Th}}{Z_{Th} + Z_L} = \frac{V_{Th}}{(R_{Th} + jX_{Th}) + (R_L + jX_L)} = \frac{V_{Th}}{(R_{Th} + R_L) + j(X_{Th} + jX_L)}$$

The average power delivered to the load is:

$$P = \frac{1}{2}|I|^2 R_L = \frac{1}{2}\frac{|V_{Th}|^2 . R_L}{(R_{Th} + R_L)^2 + (X_{Th} + X_L)^2}$$

The conditions for maximum P are $\frac{\partial P}{\partial X_L} = 0$ and $\frac{\partial P}{\partial R_L} = 0$, or:

$$\frac{\partial P}{\partial X_L} = -\left(\frac{1}{2}\right)\frac{(V_{Th}{}^2 . R_L)(2)(X_{Th} + X_L)}{\left[(R_{Th} + R_L)^2 + (X_{Th} + X_L)^2\right]^2} = 0$$

Or:

$$X_L = -X_{Th} \qquad (1)$$

From $\frac{\partial P}{\partial R_L} = 0$, we have:

$$\frac{\partial P}{\partial R_L} = \left(\frac{1}{2}\right)\frac{\left[(R_{Th} + R_L)^2 + (X_{Th} + X_L)^2\right]|V_{Th}|^2 - |V_{Th}|^2 . R_L (2)(R_{Th} + R_L)}{\left[(R_{Th} + R_L)^2 + (X_{Th} + X_L)^2\right]^2} = 0$$

Or:

$$\left[(R_{Th} + R_L)^2 + (X_{Th} + X_L)^2\right] - R_L (2)(R_{Th} + R_L) = 0$$

Or:

$$(R_{Th})^2 + (R_L)^2 + 2R_{Th} . R_L + (X_{Th} + X_L)^2 - 2R_L . R_{Th} - 2(R_L)^2 = 0$$

$$(R_{Th})^2 - (R_L)^2 + (X_{Th} + X_L)^2 = 0$$

$$(R_L)^2 - (R_{Th})^2 + (X_{Th} + X_L)^2$$

$$R_L = \sqrt{(R_{Th})^2 + (X_{Th} + X_L)^2} \qquad (2)$$

Or when the condition $X_L = -X_{Th}$ is satisfied, the second condition for maximum power transfer becomes:

$$R_L = R_{Th} \qquad (3)$$

Combining equations (1) and (3), leads to the conclusion that for maximum average power transfer, Z_L must be selected so that $X_L = -X_{Th}$ and $R_L = R_{Th}$, i.e.,

$Z_L = R_L + jX_L = R_{Th} - jX_{Th} = Z_{Th}^*$, and in this case the current in the circuit becomes:

$$I = \frac{V_{Th}}{Z_{Th} + Z_L} = \frac{V_{Th}}{(R_{Th} + jX_{Th}) + (R_L + jX_L)} = \frac{V_{Th}}{2R_{Th}}$$

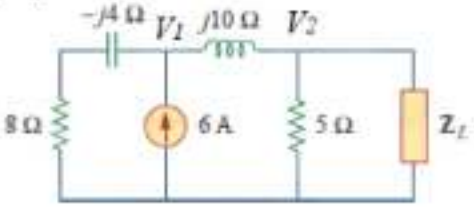
And:

$$P_{max} = \frac{1}{2}|I|^2 R_L = \frac{1}{2}\frac{|V_{Th}|^2 R_{Th}}{4(R_{Th})^2} = \frac{|V_{Th}|^2}{8R_{Th}}$$

In a situation in which the load is purely real, the condition for maximum power transfer is obtained from equation (2) by setting $X_L = 0$; that is, $R_L = \sqrt{(R_{Th})^2 + (X_{Th})^2} = |Z_{Th}|$. This means that for maximum average power transfer to a purely resistive load, the load impedance (or resistance) is equal to the magnitude of the Thevenin impedance.

Example 1:

For the circuit shown below, find the load impedance that absorbs the maximum average power. Calculate that maximum average power.



Solution:

The Thevenin's equivalent at load terminals is found. From the circuit:

$$Z_{Th} = \frac{5(8 + j6)}{13 + j6} = 3.415 + j0.732 \ \ \Omega$$

The load impedance that absorbs the maximum average power is:

$$Z_L = Z_{Th}^* = 3.415 - j0.732 \ \ \Omega$$

For this circuit, using nodal analysis:

$$\frac{V_1}{8-j4}+\frac{V_1-V_2}{j10}=6$$

$$\frac{V_2-V_1}{j10}+\frac{V_2}{5}-\frac{V_2}{3.415-j0.732}=0$$

From these 2 equations:

$$V_1=42.86+j17.14\ \ V$$

$$V_2=4.29-j8.57\ \ V$$

The current through the load is:

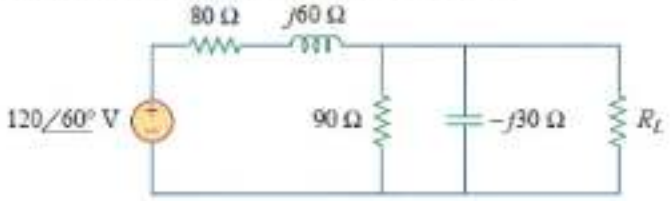
$$I=\frac{V_2}{Z_L}=\frac{4.29-j8.57}{3.415-j0.732}=1.71-j2.14=2.74\angle -51.34^{\circ}$$

And the maximum average power is:

$$P_{max}=\frac{1}{2}R_L|I|^2=\left(\frac{1}{2}\right)(3.415)(2.74)^2=12.82W$$

Example 2:

In the figure below, the resistor R_L is adjusted until it absorbs the maximum average power. Calculate R_L and the maximum average power absorbed by it.



The Thevenin's equivalent at load terminals is found. From the circuit:

$$\frac{1}{Z_{Th}}=\frac{1}{-j30}+\frac{1}{90}+\frac{1}{80+j60}=0.0191+j0.0273\ \ S$$

And:

$$Z_{Th}=17.181-j24.573\ \ \Omega$$

Thus:

$$R_L=|Z_{Th}|=\sqrt{(17.181)^2+(24.573)^2}=30\ \ \Omega$$

Using nodal analysis:

$$\frac{V-120\angle 60^{0}}{80+j60}+\frac{V}{90}+\frac{V}{-j30}+\frac{V}{30}=0$$

From which:

$$V=20.23-j1.56\ \ V$$

And the current through the load resistance is:

$$I = \frac{V}{R_t} = \frac{20.23 - j1.56}{30} = 0.674 - j0.052 \quad A = 0.676\angle -4.4^0$$

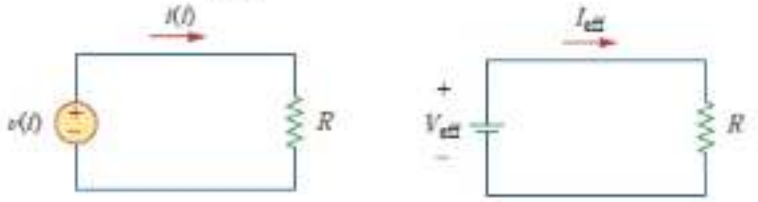
and the maximum average power absorbed by the load resistance is:

$$P_{max} = \tfrac{1}{2} R_t |I|^2 = \left(\frac{1}{2}\right)(30)(0.676)^2 = 6.86W$$

Effective or RMS Value

The effective value of a periodic current is the dc current that delivers the same average power to a resistor as the periodic current.

The figure to the left is a circuit with ac source and the circuit to right is a circuit with a dc source that produces the same average power on the resistor.



The average power absorbed by the resistor in the ac circuit is:

$$P = \frac{1}{T}\int_0^T i^2.R.dt = \frac{R}{T}\int_0^T i^2 dt \qquad (1)$$

The power absorbed by the resistor in the dc circuit is:

$$P = R.I_{eff}^2 \qquad (2)$$

From these two equations, we find:

$$I_{eff}^2 = \frac{1}{T}\int_0^T i^2 dt$$

Or:

$$I_{eff} = \sqrt{\frac{1}{T}\int_0^T i^2 dt}$$

The effective value of the voltage is found in the same way, or:

$$V_{eff} = \sqrt{\frac{1}{T}\int_0^T v^2 dt}$$

This indicates that the effective value is the (square) root of the mean (or average) of the square of the periodic signal. Thus, the effective value is often known as the root-mean-square value, or rms value for short; and we write:

$I_{eff} = I_{rms}$ and $V_{eff} = V_{rms}$.

And in general, for any periodic signal $x(t)$, the rms value is given as:

$$X_{rms} = \sqrt{\frac{1}{T}\int_0^T x^2 dt}$$

For a constant signal $x(t) = A$ is:

$$X_{rms} = \sqrt{\frac{1}{T}\int_0^T A^2 dt} = \sqrt{\frac{A^2}{T}\int_0^T dt} = \sqrt{\frac{A^2 T}{T}} = A$$

Or, the rms value of a constant is the constant itself.

For the sinusoid $i(t) = I_m.\cos\omega t$, the effective or rms value is:

$$I_{rms} = \sqrt{\frac{1}{T}\int_0^T (I_m \cos\omega t)^2\, dt} = \sqrt{\frac{I_m^2}{T}\int_0^T \left(\frac{1+\cos 2\omega t}{2}\right) dt}$$

$$= I_m \sqrt{\frac{1}{2T}\left[\int_0^T (1-\cos 2\omega t)dt\right]} = I_m \sqrt{\frac{1}{2T}\left[\int_0^T dt - \int_0^T \cos 2\omega t\, dt\right]}$$

$$= I_m \sqrt{\frac{1}{2T}T} = \frac{I_m}{\sqrt{2}}$$

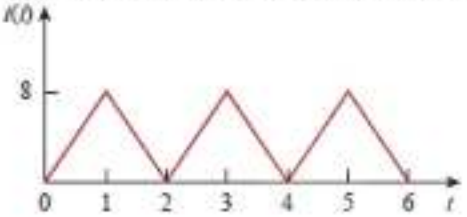
Similarly, for $v(t) = V_m.\cos\omega t$, $V_{rms} = \frac{V_m}{\sqrt{2}}$.

The average power was found to be:

$$P = \frac{1}{2}V_m.I_m.\cos(\theta_v - \theta_i) = \frac{V_m}{\sqrt{2}}.\frac{I_m}{\sqrt{2}}.\cos(\theta_v - \theta_i) = V_{rms}.I_{rms}.\cos(\theta_v - \theta_i)$$ Find the rms value of the

Example:

Find the rms value for the current waveform shown in the figure below. If this current flows through a 9Ω resistor, calculate the average power absorbed by this resistor.



Solution:

This current waveform is defined as:

$$i(t) = \begin{cases} 8t & for \quad 0 \le t \le 1 \\ -8t+16 & for \quad 1 \le t \le 2 \end{cases} \text{ and } \quad T = 2$$

$$I_{rms} = \sqrt{\frac{1}{T}\int_0^T i^2 dt} = \sqrt{\frac{1}{2}\left[\int_0^1 (8t)^2\, dt + \int_1^2 (-8t+16)^2\, dt\right]}$$

$$= \sqrt{\frac{1}{2}\int_0^1 64t^2 dt + \frac{1}{2}\int_1^2 (64t^2 - 256t + 256)dt} = \sqrt{\frac{64}{6}t^3\Big|_0^1 - \frac{64t^3}{6}\Big|_1^2 - \frac{256t^2}{4}\Big|_1^2 + \frac{256t}{2}\Big|_1^2}$$

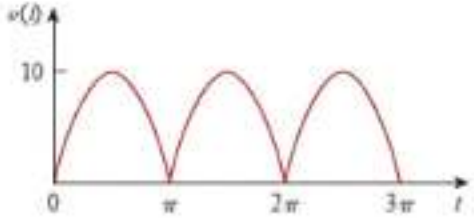
$$= \sqrt{\frac{64}{6} + \frac{64}{6}(7) - \frac{256}{4}(3) + \frac{256}{2}(1)} = 4.619A$$

And:

$$P = R.I_{rms}^2 = 9(4.619)^2 = 192W$$

Example:

Find the rms value of the full-wave rectified sine wave in the figure below. Calculate the average power dissipated in a 6Ω resistor.



Solution:

For this waveform:

$v(t) = 10\sin \omega t$ and $T = \pi$

$$V_{rms} = \sqrt{\frac{1}{T}\int_0^T v^2 dt} = \sqrt{\frac{1}{\pi}\int_0^\pi 100\sin^2 \omega t.dt} = \sqrt{\frac{100}{2\pi}\int_0^\pi (1-\cos 2\omega t)dt}$$

$$= \sqrt{\frac{100}{2\pi}\left[\int_0^\pi dt - \int_0^\pi \cos 2\omega t.dt\right]} = \sqrt{\frac{100\pi}{2\pi}} = 7.071V$$

$$P = \frac{V_{rms}^2}{R} = \frac{(7.071)^2}{6} = 8.33W$$

Apparent Power and Power Factor

We have found previously that the average power is:

$$P = \frac{1}{2}V_m.I_m.\cos(\theta_v - \theta_i) = V_{rms}.I_{rms}.\cos(\theta_v - \theta_i)$$

The product $V_{rms}.I_{rms}$ is called the apparent power and usually is denoted by S, while $\cos(\theta_v - \theta_i)$ is called the power factor (pf), or:

$P = S.pf$

Where:

$S = V_{rms}.I_{rms}$

And:

$pf = \cos(\theta_v - \theta_i)$

And the angle $(\theta_v - \theta_i)$ is called the power factor angle.

Note that the power factor angle is equal to the angle of the load impedance, this is clear as:

$$Z = \frac{V}{I} = \frac{V_m\angle\theta_v}{I_m\angle\theta_i} = \frac{V_m}{I_m}\angle(\theta_v - \theta_i)$$

This means that, the power factor is the cosine of the phase difference between voltage and current. And is also the cosine of the angle of the load impedance.

Note that the apparent power is measured in volt-amperes or VA while the average or real power is measured in watts. The power factor is dimensionless.

The value of *pf* ranges between zero and unity. For a purely resistive load, the voltage and current are in phase, so that $\theta_v - \theta_i = 0$ and $pf = \cos(\theta_v - \theta_i) = 1$. This implies that the apparent power is equal to the average power. For a purely reactive load, $\theta_v - \theta_i = 90^0$ and $pf = \cos(\theta_v - \theta_i) = 0$. In this case the average power is zero. In between these two extreme cases, pf is said to be leading or lagging. Leading power factor means that current leads voltage, which implies a capacitive load. Lagging power factor means that current lags voltage, implying an inductive load.

Example:
Obtain the power factor and the apparent power of a load whose impedance is $Z = 60 + j40\ \Omega$ when the applied voltage is $v(t) = 160\cos(377t + 10^0)\ V$

Solution:

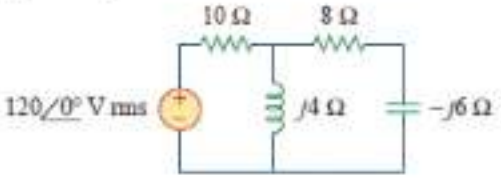
$$\theta = \tan^{-1}\left(\frac{40}{60}\right) = 33.69^0$$

$pf = \cos\theta = 0.832$ lagging (as the load is inductive).

$$I = \frac{V}{Z} = \frac{160\angle 10^0}{60 - j40} = \frac{160\angle 10^0}{72.11\angle 33.69^0} = 2.219\angle -23.69^0$$

$$S = V_{rms} I_{rms} = \left(\frac{160}{\sqrt{2}}\right)\left(\frac{2.219}{\sqrt{2}}\right) = 177.52\ \ VA$$

Example:
Calculate the power factor of the entire circuit in the following figure as seen by the source. What is the average power supplied by the source?



Solution:

$$Z = 10 + j4 // (8 - j6) = 10 + \frac{j4(8 - j6)}{8 - j2} = 11.88 + j4.47\ \ \Omega = 12.7\angle 20.62^0$$

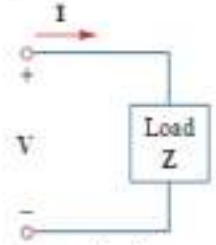
$pf = \cos 20.62^0 = 0.936$ *lagging*

$$I = \frac{V}{Z} = \frac{120\angle 0^0}{12.7\angle 20.62^0} = 9.45\angle -20.62^0$$

$$P = V_{rms} I_{rms} pf = (120)(9.45)(0.936) = 1062\ \ W = 1.062\ \ kW$$

Complex Power

Consider the ac load in the figure below.



The complex power S absorbed by the ac load is the product of the voltage and the complex conjugate of the current, or:

$$S = \frac{1}{2} V.I^* = V_{rms} I^*_{rms} = |V_{rms}| \angle \theta_v . |I_{rms}| \angle -\theta_i$$

$$= |V_{rms}| |I_{rms}| \angle (\theta_v - \theta_i) = |V_{rms}| |I_{rms}| \cos(\theta_v - \theta_i) + j |V_{rms}| |I_{rms}| \sin(\theta_v - \theta_i)$$

$$= P + jQ$$

Or:

$P = \mathrm{Re}(S) = |V_{rms}| |I_{rms}| \cos(\theta_v - \theta_i)$ is the active power, measured in Watts or (W).and:

$Q = \mathrm{Im}(S) = |V_{rms}| |I_{rms}| \sin(\theta_v - \theta_i)$ is the reactive power, measured in VA reactive, or (VAR), and:

$|S| = \sqrt{P^2 + Q^2} = |V_{rms}| |I_{rms}|$ is the apparent power, measured in VA, or (VA).

Note that, the real power P is the average power in watts delivered to a load; it is the only useful power. It is the actual power dissipated by the load. The reactive power Q is a measure of the energy exchange between the source and the reactive part of the load. The unit of Q is the volt-ampere reactive (VAR).

We know that the capacitor and the inductor are energy storage elements as they neither dissipate nor supply power, but exchange power back and forth with the rest of the network. In the same way, the reactive power is being transferred back and forth between the load and the source. It represents a lossless interchange between the load and the source. Notice that:

- for resistive loads (unity pf), $Q = 0$,
- for capacitive loads (leading pf), $Q < 0$
- for inductive loads (lagging pf), $Q > 0$

The complex power may be expressed in terms of the load impedance Z, as:

$$Z = \frac{V}{I} = \frac{|V| \angle \theta_v}{|I| \angle \theta_i} = \frac{|V|}{|I|} \angle (\theta_v - \theta_i) = R + jX$$

Also:

$V = I.Z$, thus:

$$S = V.I^{*} = Z.I_{rms}.I_{rms}^{*} = Z\left|I_{rms}\right|^{2} = \left|I_{rms}\right|^{2}(R + jX)$$

$$= R\left|I_{rms}\right|^{2} + jX\left|I_{rms}\right|^{2} = P + jQ$$

Or:

$$P = R\left|I_{rms}\right|^{2}$$

$$Q = X\left|I_{rms}\right|^{2}$$

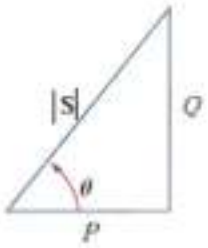
Also as:

$I = \frac{V_{rms}}{Z}$, thus: $I^{*} = \frac{V_{rms}^{*}}{Z^{*}}$

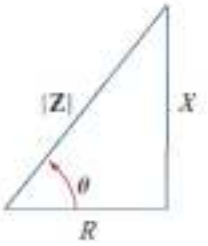
And, as:

$$S = V_{rms}.I_{rms}^{*} = V_{rms}.\frac{V_{rms}^{*}}{Z^{*}} = \frac{\left|V_{rms}\right|^{2}}{Z^{*}}$$

It is a standard practice to represent S, P, and Q in the form of a triangle, known as the power triangle, as shown in the figure below.

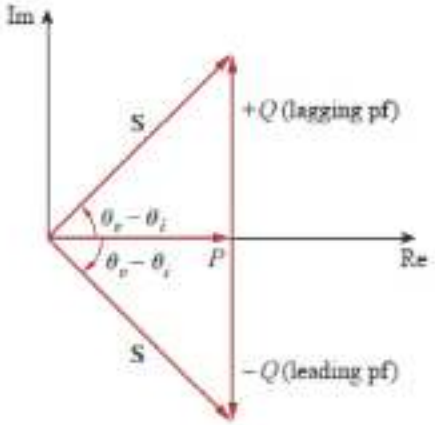


This triangle is similar to the impedance triangle showing the relationship between Z, R, and X, illustrated in following figure.



Note that the power triangle has four items—the apparent/complex power, real power, reactive power, and the power factor angle. Given two of these items, the other two can easily be obtained from the triangle.

As in the following figure, when S lies in the first quadrant, we have an inductive load and a lagging *pf*. When S lies in the fourth quadrant, the load is capacitive and the *pf* is leading.



It is also possible for the complex power to lie in the second or third quadrant. This requires that the load impedance have a negative resistance, which is possible with active circuits, (circuits with power sources).

Example 1:

For a load, $V_{rms} = 110\angle 85^{\circ} V$, $I_{rms} = 0.4\angle 15^{\circ} A$ Determine: (a) the complex and apparent powers, (b) the real and reactive powers, and (c) the power factor and the load impedance.

Solution:

a) $S = V_{rms} I_{rms}^{*} = (110\angle 85^{0})(0.4\angle -15^{0}) = 44\angle 70^{0} VA$

$|S| = V_{rms} I_{rms}^{*} = 44VA$

b) $S = 44\angle 70^{0} = 44(\cos 70^{0} + j \sin 70^{0}) = 15.05 + j41.35 = P + jQ$, or:

$P = 15.05W$ and $Q = 41.35VAR$

c) $pf = \cos(70^{0}) = 0.342$ lagging

$$Z = \frac{V_{rms}}{I_{rms}} = \frac{110\angle 85^{\circ}}{0.4\angle 15^{\circ}} = 275\angle 70^{0} = 94.06 + j285.42 \ \ \Omega$$

Example:

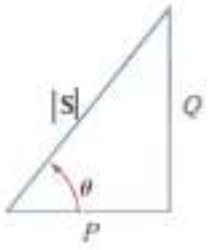
A sinusoidal source supplies $20kVAR$ reactive power to load $Z = 250\angle -75^{0}$.

Determine: (a) the power factor, (b) the apparent power delivered to the load, and (c) the rms voltage.

Solution:

a) $pf = \cos(75^{\circ}) = 0.2588$ leading.

b) from the power triangle shown below.



$$|S| = \frac{Q}{\sin\theta} = \frac{20x10^3}{\sin 75^\circ} = 20.706 \quad kVA$$

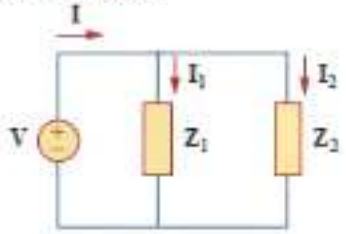
c) $S = |S|(\cos\theta + j\sin\theta) = 20706(\cos 75^\circ - j\sin 75^\circ) = 5359 + j20000 \quad VA$

but $S = \frac{|V_{rms}|^2}{Z^*}$, or:

$$|V_{rms}| = \sqrt{S.Z^*} = \sqrt{(5359 + j20000)(250\angle 75^\circ)} = |588.86 + j2197.6| = 2.275kV$$

Conservation of AC Power

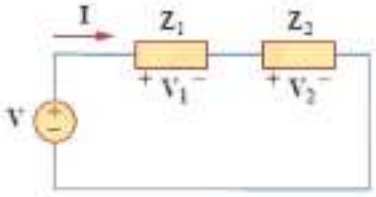
The principle of conservation of power applies to ac circuits as well as to dc circuits. Consider a circuit with two parallel loads, as shown below.



For this circuit:

$S = VI^* = V(I_1^* + I_2^*) = VI_1^* + VI_2^* = S_1 + S_2$

Also for circuit with series loads, as below:



$S = VI^* = (V_1 + V_2)I^* = V_1I^* + V_2I^* = S_1 + S_2$

Or, whether the loads are connected in series or in parallel (or in general), the total power supplied by the source equals the total power delivered to the load.

Also, as:

$S = P + jQ = P_1 + jQ_1 + P_2 + jQ_2 = (P_1 + P_2) + j(Q_1 + Q_2)$

Or:

$P = P_1 + P_2$, and:

$Q = Q_1 + Q_2$

This means that total active (or reactive) power supplied by the source equals the total active (or reactive) power delivered to the load.

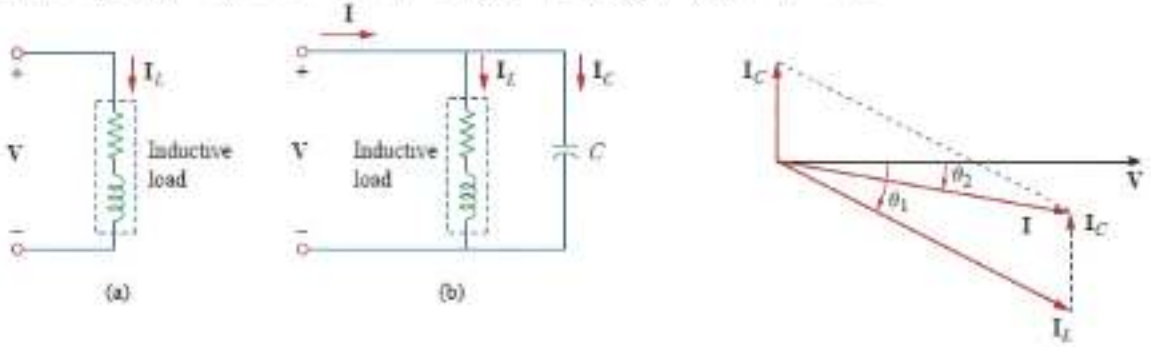
Power Factor Correction

Most domestic loads (such as washing machines, air conditioners, and refrigerators) and industrial loads (such as induction motors) are inductive and operate at a low lagging power factor.

Power factor correction is the addition of a reactive element (usually a capacitor) in parallel with the load in order to make the power factor closer to unity.

Figure (a) below shows a circuit that has a power factor of $\cos\theta_1$ while the one in (b) has a power factor of $\cos\theta_2$. It is evident from the phasor diagram that adding the capacitor has caused the phase angle between the supplied voltage and current to reduce from θ_1 to θ_2, thereby increasing the power factor. We also notice from the magnitudes of the vectors in phasor diagram that with the same supplied voltage, the circuit in figure (a) draws larger current I_L than the current I drawn by the circuit in figure (b).

Power companies charge more for larger currents, because they result in increased power losses (by a squared factor, since $P = I_L^2 . R$). Therefore, it is beneficial to both the power company and the consumer that every effort is made to minimize current level or keep the power factor as close to unity as possible. By choosing a suitable size for the capacitor, the current can be made to be completely in phase with the voltage, implying unity power factor



Consider the power triangle in the figure below. If the original inductive load has apparent power S_1, then $P = S_1 . \cos\theta_1$, $\quad Q_1 = S_1 . \sin\theta_1 = P\tan\theta_1$

If we desire to increase the power factor from $\cos\theta_1$ to $\cos\theta_2$, the new reactive power is:

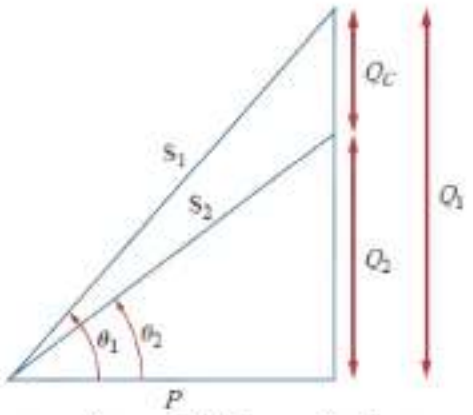
$Q_2 = P\tan\theta_2$

The reduction in the reactive power is caused by the shunt capacitor; that is:

$Q_C = Q_1 - Q_2 = P(\tan\theta_1 - \tan\theta_2)$ $\quad$ (1)

But:

$$Q_C = \frac{V_{rms}^2}{X_C} = \omega C V_{rms}^2 \qquad (2)$$



The value of the required shunt capacitance C is determined as:

$$C = \frac{Q_C}{\omega V_{rms}^2} = \frac{P(\tan\theta_1 - \tan\theta_2)}{\omega V_{rms}^2} \qquad (3)$$

Note: Although the most common situation in practice is that of an inductive load, it is also possible that the load is capacitive; that is, the load is operating at a leading power factor. In this case, an inductor should be connected across the load for power factor correction. The required shunt inductance L can be calculated from:

$Q_L = \frac{V_{rms}^2}{X_L} = \frac{V_{rms}^2}{\omega L}$ or $L = \frac{V_{rms}^2}{\omega Q_L}$, where $Q_L = Q_1 - Q_2$, the difference between the new and old reactive powers.

Example:

Find the value of parallel capacitance needed to correct a load of $140kVAR$ at 0.85 lagging pf to unity pf. Assume that the load is supplied by a 110-V (rms), 60-Hz line

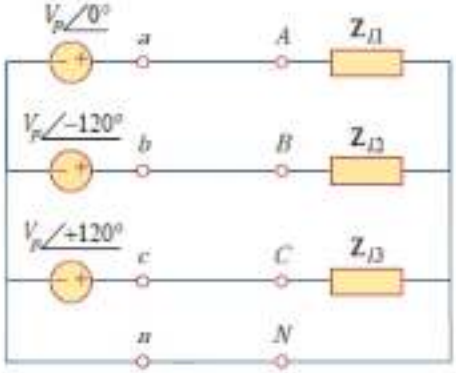
Solution:

As the pf becomes unity, the net reactive power of the load and the power-factor correction capacitor becomes zero. Or the reactive power generated by the capacitor must be equal to the reactive power consumed by the load, or, $Q_C = 140kVAR$, thus:

$$C = \frac{Q_C}{\omega V_{rms}^2} = \frac{140x10^3}{2\pi(60)(110^2)} = 0.0307F = 30.7mF$$

Three-Phase Circuits

The figure below shows a three-phase four-wire system.



This system is produced by a generator consisting of three sources having the same amplitude and frequency but out of phase with each other by 120°.

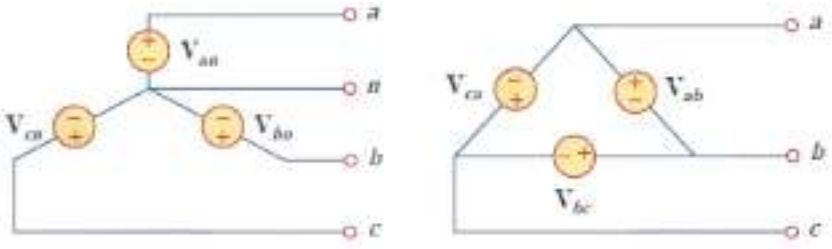
Three-phase systems are important for at least three reasons:
1) Nearly all electric power is generated and distributed in three-phase, at the operating frequency of 50 Hz (or in 60Hz. in the United States).

2) The instantaneous power in a three-phase system can be constant (not pulsating). This results in uniform power transmission and less vibration of three-phase machines.

3) For the same amount of power, the three-phase system is more economical than the single-phase. The amount of wire required for a three-phase system is less than that required for an equivalent single-phase system.

Balanced Three-Phase Voltages

A typical three-phase system consists of three voltage sources connected to loads by three or four wires (or transmission lines). A three-phase system is equivalent to three single-phase circuits. The voltage sources can be either wye-connected as shown in figure (a) or delta-connected as in figure (b).



(a) (b)

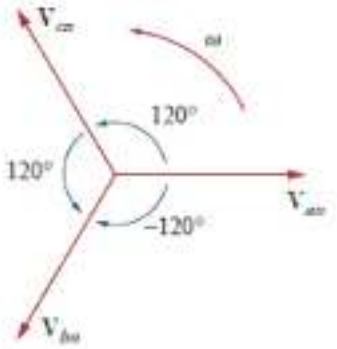
For the wye-connected voltages in figure (a), the voltages and are respectively between lines a, b, and c, and the neutral line n. These voltages are called phase voltages. Usually the three voltage

sources have the same amplitude and frequency and are out of phase with each other by 120^0, the voltages are said to be balanced. This implies that:

$V_{an} + V_{bn} + V_{cn} = 0$

Since the three-phase voltages are 120^c out of phase with each other, there are two possible combinations:

1) Positive-sequence (or *abc* sequence) system. in this system, V_{bn} lags V_{an} and V_{cn} leads V_{an}, as shown below.



Or mathematically:

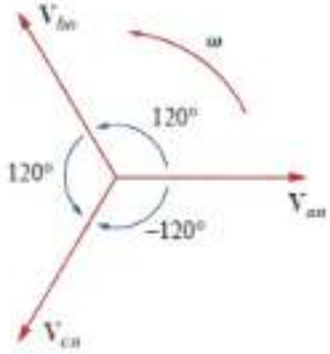
$V_{an} = V_p \angle 0^0$

$V_{bn} = V_p \angle -120^0$

$V_{cn} = V_p \angle 120^0$

Where V_p is the effective or rms value of the phase voltages.

2) Negative-sequence (or *acb* sequence) system. in this system, V_{bn} leads V_{an} and V_{cn} lags V_{an}, as shown below.



Or mathematically:

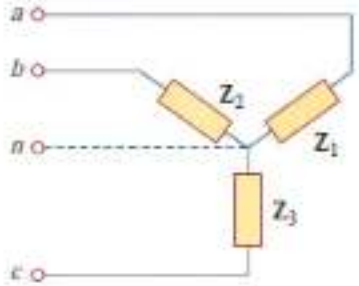
$V_{an} = V_p \angle 0^0$

$V_{bn} = V_p \angle 120^0$

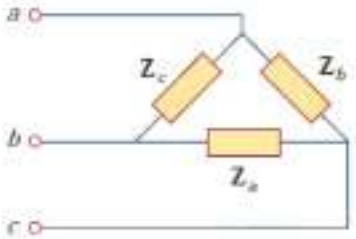
$V_{cn} = V_p \angle -120^0$

Note that, the phase sequence is the time order in which the voltages pass through their respective maximum values.

Like the generator connections, a three-phase load can be either wye-connected or delta-connected, as shown below.



wye-connected load



delta-connected load

The neutral line in the wye-connected load may or may not be there, depending on whether the system is four- or three-wire.

Note that, A balanced load is one in which the phase impedances are equal in magnitude and in phase. Or, for a balanced wye-connected load,

$Z_1 = Z_2 = Z_3 = Z_Y$

where Z_Y is the load impedance per phase. For a balanced delta-connected load,

$Z_a = Z_b = Z_c = Z_\Delta$

where Z_Δ is the load impedance per phase.

Note that, as found before:

$Z_\Delta = 3Z_Y$ or $Z_Y = \frac{1}{3}Z_\Delta$

Since both the three-phase source and the three-phase load can be either wye- or delta-connected, we have four possible connections:

1) $Y - Y$ connection (i.e., Y-connected source with a Y-connected load).
2) $Y - \Delta$ connection.
3) $\Delta - \Delta$ connection.
4) $\Delta - Y$ connection.

Example:

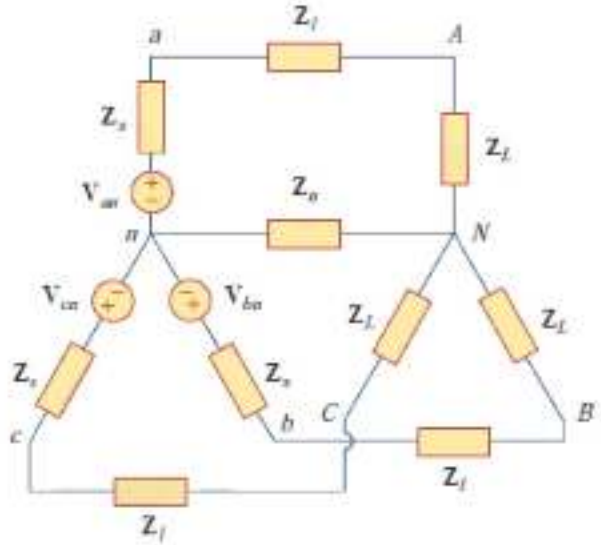
Given $V_{ab} = 110\angle 30^0$, find V_{an} and V_{cn} assuming a positive (abc) sequence.

Solution:

For the positive sequence system, V_{an} leads V_{bn} by 120^0, or $V_{an} = 110\angle 150^0$ and V_{cn} lags V_{bn} by 120^0, or

Balanced Wye-Wye Connection

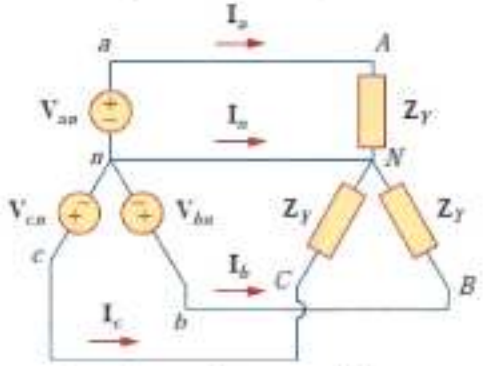
A balanced Y-Y system is a three-phase system with a balanced Y-connected source and a balanced Y-connected load. See the figure below.



In this figure, Z_S denotes the internal impedance of the phase winding of the generator; Z_l is the impedance of the line joining a phase of the source with a phase of the load; Z_L is the impedance of each phase of the load; and Z_n is the impedance of the neutral line. Thus, in general:

$$Z_Y = Z_S + Z_l + Z_L$$

And the Y-Y system in the previous figure can be simplified to that shown in following figure.



Assuming the positive sequence, the phase voltages (or line-to-neutral voltages) are:

$$V_{an} = V_p \angle 0^\circ, \quad V_{bn} = V_p \angle -120^0 \quad \text{and} \quad V_{cn} = V_p \angle 120^0$$

The line-to-line voltages or simply line voltages and are:

$$V_{ab} = V_a - V_b = V_p \angle 0^0 - V_p \angle -120^0 = V_p\left(1 - 1\angle -120^\circ\right)$$

$$= V_p\left(1 - \cos 120^\circ + j \sin 120^0\right) = V_p\left(1 + \frac{1}{2} + j\frac{\sqrt{3}}{2}\right)$$

$$= V_p\left(1 + \frac{1}{2} + j\frac{\sqrt{3}}{2}\right) = V_p\left(\frac{3}{2} + j\frac{\sqrt{3}}{2}\right) = V_p\sqrt{\left(\frac{3}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2}\angle \tan^{-1}\left(\frac{\frac{\sqrt{3}}{2}}{\frac{3}{2}}\right)$$

$$= V_p\sqrt{3}\angle \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \sqrt{3}V_p\angle 30^0$$

In the same way:

$V_{bc} = V_b - V_c = \sqrt{3}V_p\angle -90^0$

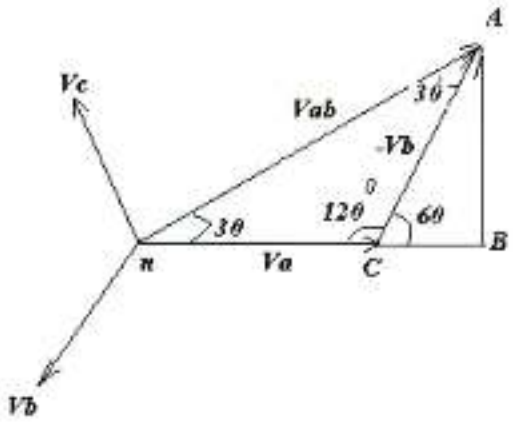
$V_{ca} = V_c - V_a = \sqrt{3}V_p\angle 150^0$

Thus, the magnitude of the line voltages is times the magnitude of the phase voltages or:

$V_L = \sqrt{3}V_p$

Note also that the line voltage $V_{ab} = V_a - V_b$ can be found graphically as shown below.

from the figure below:



$|V_{ab}|.\sin 30^0 = |V_a|.\sin 60^0 = AB$

or:

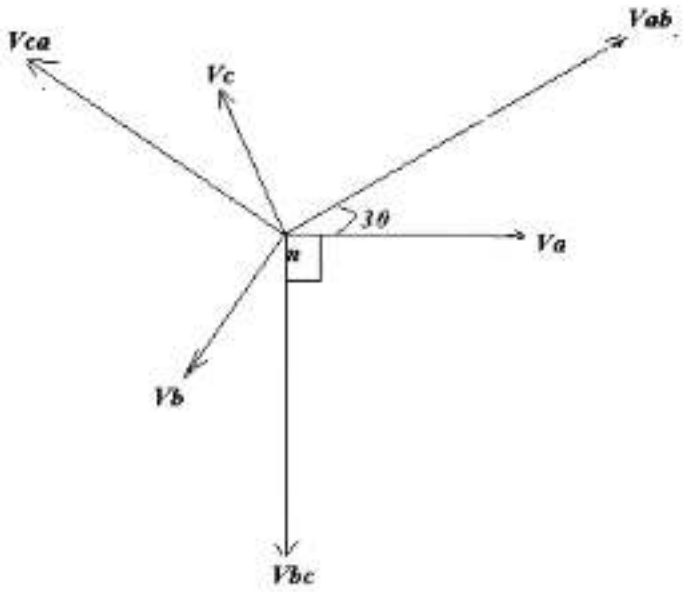
$$\frac{1}{2}V_{ab}| = \frac{\sqrt{3}}{2}|V_a|$$

or:

$|V_{ab} = \sqrt{3}|V_a|$ and is shifted 30^0

or: $V_{ab} = \sqrt{3}|V_a|\angle 30^0$

The other line voltages are found in a similar manner and are shown in the figure below..



Note that the sum of the three phase voltages in the balanced three-phase system is zero, or:

$$V_a + V_b + V_c = V_p\angle 0^0 + V_p\angle -120^0 + V_p\angle 120^0 = 0$$

Also, the sum of the three line voltages in the three-phase system (balanced or not) is zero, or:

$$V_{ab} - V_{bc} + V_{ca} = V_a - V_b + V_b - V_c + V_c - V_a = 0$$

Applying KVL to each phase in the balanced three-phase system, we obtain the line currents as:

$$I_a = \frac{V_{an}}{Z_Y}, \quad I_b = \frac{V_{bn}}{Z_Y} = \frac{V_{an}\angle -120^0}{Z_Y} = I_a\angle -120^0 \text{ and } I_c = \frac{V_{cn}}{Z_Y} = \frac{V_{an}\angle 120^0}{Z_Y} = I_a\angle 120^0$$

It can be seen that:

$$I_a + I_b + I_c = 0$$

And:

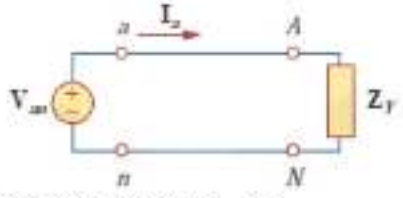
$$I_n = -(I_a + I_b + I_c) = 0$$

Or:

$$V_{nN} = Z_n I_n = 0$$

That is, the voltage across the neutral wire is zero. The neutral line can thus be removed without affecting the system.

As the currents in the balanced three-phase system have equal magnitudes but phase shifted by 120^0, an alternative way of analyzing a balanced Y-Y system is to do so on a "per phase" basis. We look at one phase, say phase a, and analyze the single-phase equivalent circuit, as shown below.



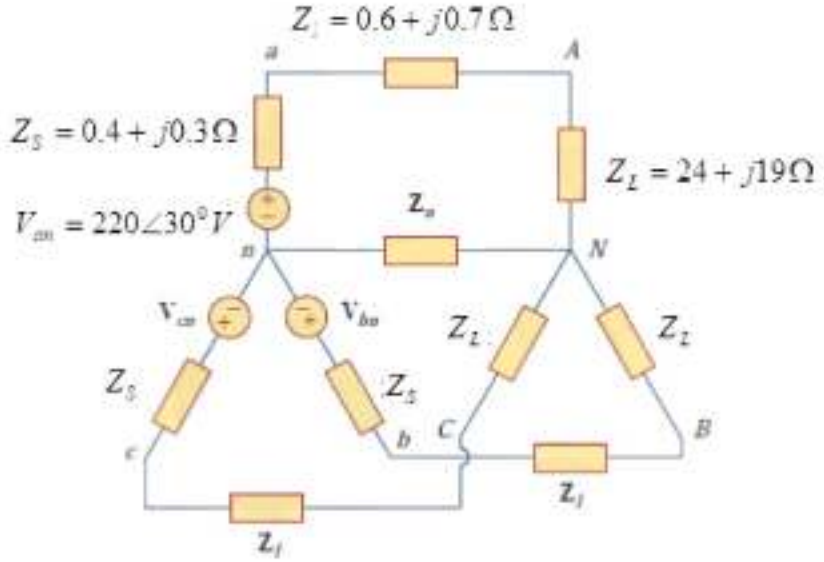
The single-phase analysis yields the line current I_a as:

$$I_a = \frac{V_{an}}{Z_Y}$$

From I_a we use the phase sequence to obtain other line currents. Thus, as long as the system is balanced, we need only analyze one phase. We may do this even if the neutral line is absent, as in the three-wire system.

Example:

A Y-connected balanced three-phase generator with an impedance of $Z_s = 0.4 + j0.3\Omega$ per phase is connected to a Y-connected balanced load with an impedance of $Z_L = 24 + j19\Omega$ per phase. The line joining the generator and the load has an impedance of $Z_l = 0.6 + j0.7\Omega$ per phase. Assuming a positive sequence for the source voltages and that $V_{an} = 220\angle 30^\circ V$, see the figure below.



find:

a) all the phase and the line voltages,

b) the phase and the line currents,

c) the current in the neutral wire.

Solution:

a) $V_{an} = 220\angle 30^\circ V$, $V_{bn} = 220\angle -90^\circ V$ and $V_{cn} = 220\angle 150^\circ V$

$V_{ab} = 381\angle 60^\circ V$, $V_{bc} = 381\angle -60^\circ V$ and $V_{ca} = 381\angle 180^\circ V$

b) The impedance in the phase circuit is:

$$Z_Y = Z_s + Z_l + Z_L = (0.4 + j0.3) + (24 + j19) + (0.6 + j0.7) = 25 + j20\Omega$$

The respective line and phase currents are:

$$I_a = \frac{V_{an}}{Z_Y} = \frac{220\angle 30^\circ}{25 + j20} = \frac{220\angle 30^\circ}{32.02\angle 38.66^\circ} = 6.87\angle -8.66^\circ A$$

$$I_a = \frac{V_{an}}{Z_Y} = 6.87\angle -128.66^0\, A$$

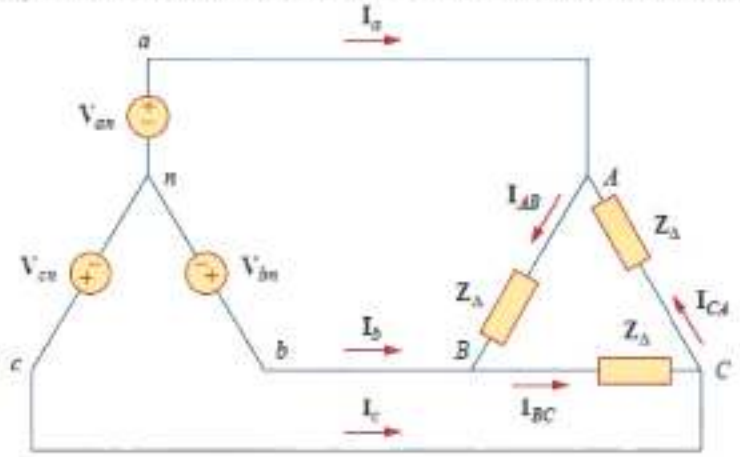
$$I_c = \frac{V_{cn}}{Z_Y} = 6.87\angle 111.34^0\, A$$

c) $I_a + I_b + I_c = 0$

Three-Phase Circuits

Balanced Wye-Delta Connection

Here, the source is Y-connected and the load is Δ -connected. See the figure below.



Assuming the positive sequence, the phase voltages are:

$V_{an} = V_p \angle 0^\circ$, $V_{bn} = V_p \angle -120^\circ$ and $V_{cn} = V_p \angle 120^\circ$

And the line voltages are:

$$V_{ab} = \sqrt{3} V_p \angle 30^\circ = V_{AB}, \quad V_{bc} = \sqrt{3} V_p \angle -90^\circ = V_{BC}, \quad V_{ca} = \sqrt{3} V_p \angle 150^\circ = V_{CA}$$

Note that the line voltages are equal to the voltages across the load impedances and the phase currents are;

$$I_{AB} = \frac{V_{AB}}{Z_\Delta}, \quad I_{BC} = \frac{V_{BC}}{Z_\Delta} \quad \text{and} \quad I_{CA} = \frac{V_{CA}}{Z_\Delta}$$

Note that these currents have the same magnitude but are out of phase with each other by -120°

The line currents are obtained from the phase currents by applying KCL at nodes A, B, and C. Thus;

$$I_a = I_{AB} - I_{CA} = I_{AB} - I_{AB} \angle 120^\circ = I_{AB} \left[1 - \left(\cos 120^\circ + j \sin 120^\circ \right) \right]$$

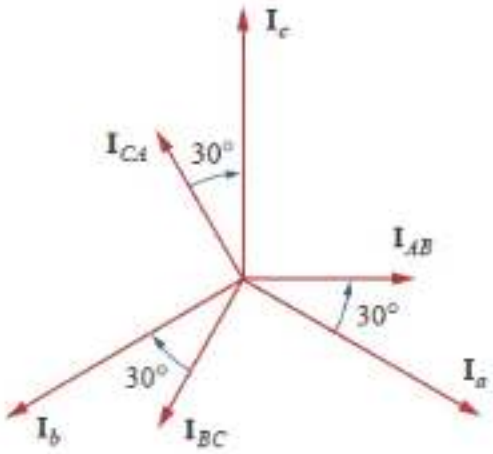
$$= I_{AB} \left[1 - \left(-\frac{1}{2} + j \frac{\sqrt{3}}{2} \right) \right] = I_{AB} \left(\frac{3}{2} - j \frac{\sqrt{3}}{2} \right) = \sqrt{3} I_{AB} \angle -30^\circ$$

Note that the magnitude of the line current is $\sqrt{3}$ times the magnitude of the phase current, or:

$I_L = \sqrt{3} I_p$

Also, the line currents lag the corresponding phase currents by 30°, assuming the positive sequence system

The figure below shows the phasor diagram illustrating the relationship between the phase and line currents.



An alternative way of analyzing the $Y-\Delta$ circuit is to transform the Δ-connected load to an equivalent Y-connected load. Using:

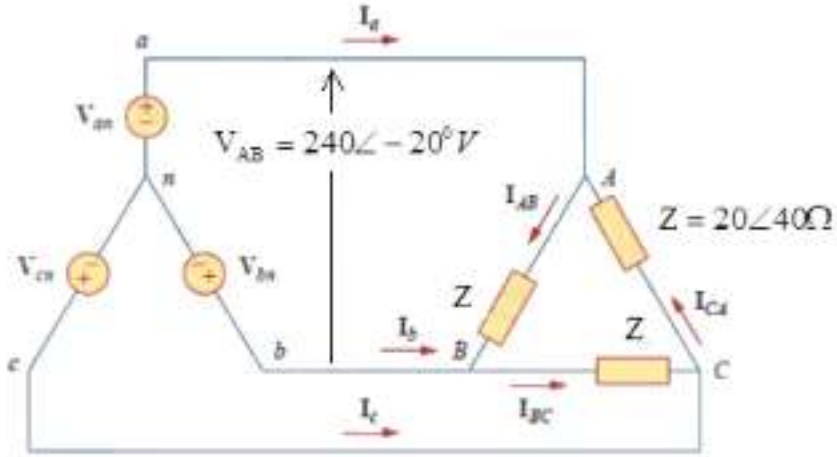
$$Z_Y = \frac{Z_\Delta}{3}$$

And then the three-phase can be replaced by the single-phase equivalent circuit. This allows us to calculate only the line currents. The phase currents are found utilizing the fact that each of the phase currents leads the corresponding line current by 30^0.

Example;

In a positive-sequence (abc) system, the line voltage V_{AB} is $V_{AB} = 240\angle -20^0 V$. If the source is connected to a Δ-connected load of $Z = 20\angle 40\Omega$, find the phase and line currents.

Solution:



The phase currents are;

$$I_{AB} = \frac{V_{AB}}{Z} = \frac{240\angle -20^0}{20\angle 40^0} = 12\angle -60^0 A$$

$$I_{BC} = I_{AB}\angle -120^0 = 12\angle -180^0 A$$

$$I_{CA} = I_{AB}\angle 120^0 = 12\angle 60^0 A$$

The line currents are;

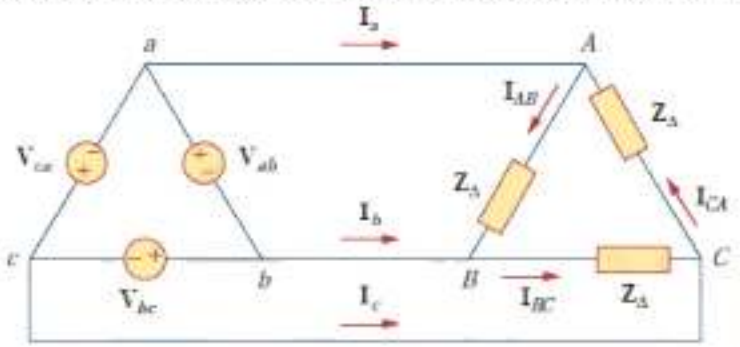
$$I_a = \sqrt{3} I_{AB} \angle -30^{\circ} = 12\sqrt{3} \angle -90^{\circ} A$$

$$I_b = I_a \angle -120 = 12\sqrt{3} \angle -210^{\circ} A$$

$$I_c = I_a \angle 120 = 12\sqrt{3} \angle 30^{\circ} A$$

Balanced Delta-Delta Connection

In this circuit the source as well as the load are delta-connected as shown below.



Assuming a positive sequence, the phase voltages for a delta-connected source are:

$V_{ab} = V_p \angle 0^{\circ}$, $V_{bc} = V_p \angle -120^{\circ}$ and $V_{ca} = V_p \angle 120^{\circ}$

Note that the line voltages are the same as the phase voltages and the phase voltages of the delta-connected source are equal to the voltages across the impedances, that is:

$V_{ab} = V_{AB}$, $V_{bc} = V_{BC}$ and $V_{ca} = V_{CA}$.

Hence, the phase currents are:

$$I_{AB} = \frac{V_{AB}}{Z_\Delta} = \frac{V_{ab}}{Z_\Delta}, \quad I_{BC} = \frac{V_{BC}}{Z_\Delta} = \frac{V_{bc}}{Z_\Delta} \quad \text{and} \quad I_{CA} = \frac{V_{CA}}{Z_\Delta} = \frac{V_{ca}}{Z_\Delta}$$

The line currents are:

$$I_a = I_{AB} - I_{CA} = I_p \angle 0^{\circ} - I_p \angle 120^{\circ} = I_p \left[1 - \left(\cos 120^{\circ} + j \sin 120^{\circ}\right)\right]$$

$$I_p \left[1 - \left(-\frac{1}{2} + j\frac{\sqrt{3}}{2}\right)\right] = I_p \left(\frac{3}{2} - j\frac{\sqrt{3}}{2}\right) = \sqrt{3} I_p \angle -30^{\circ}$$

In the same way:

$$I_b = I_{BC} - I_{AB} = \sqrt{3} I_p \angle -150^{\circ}$$

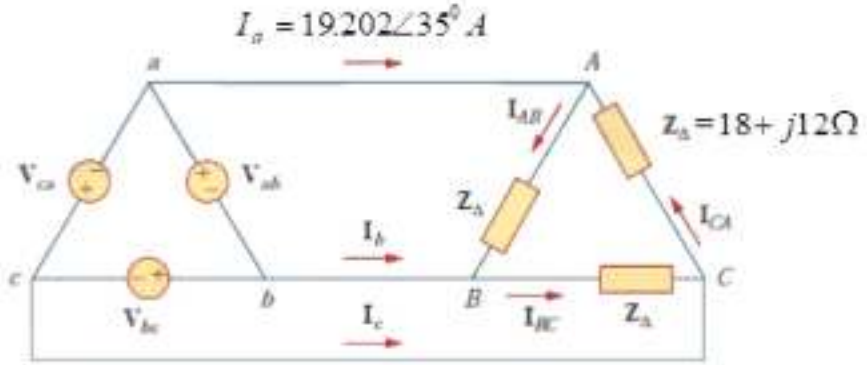
$$I_c = I_{CA} - I_{BC} = \sqrt{3} I_p \angle 90^{\circ}$$

Note: An alternative way of analyzing the $\Delta - \Delta$ circuit is to convert both the source and the load to their Y equivalents.

Example:

A positive-sequence, balanced Δ-connected source supplies a balanced Δ-connected load. If the impedance per phase of the load is $18 + j12\Omega$ and $I_a = 19.202 \angle 35^{\circ} A$, find I_{AB} and V_{AB}.

Solution:



$$I_a = I_{AB} - I_{CA} = I_P\angle 0^\circ - I_P\angle 120^\circ = I_P\left[1 - \left(\cos 120^\circ + j\sin 120^\circ\right)\right]$$

$$I_P\left[1 - \left(-\frac{1}{2} + j\frac{\sqrt{3}}{2}\right)\right] = I_P\left(\frac{3}{2} - j\frac{\sqrt{3}}{2}\right) = \sqrt{3}I_P\angle -30^\circ$$

Or:

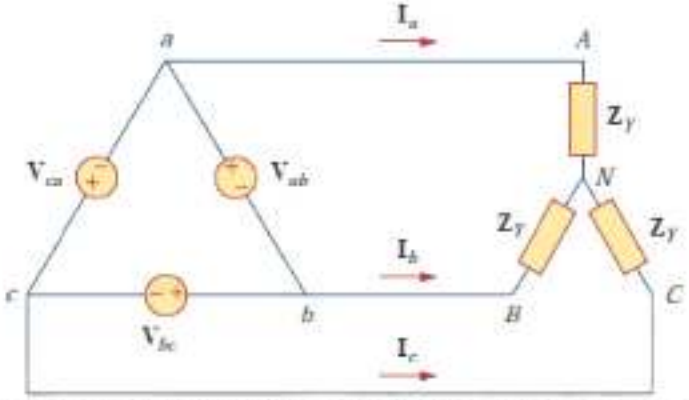
$$I_{AB} = \frac{19.202\angle 35^\circ}{\sqrt{3}\angle -30^\circ} = 11.09\angle 65^\circ A$$

$$V_{AB} = I_{AB}.Z_\Delta = 11.09\angle 65^\circ(18 + j12) = \left(11.09\angle 65^\circ\right)\left(21.63\angle 33.69^\circ\right) = 239.88\angle 98.69^\circ V$$

Three-Phase Circuits

Balanced Delta-Wye Connection

This system has a balanced Δ-connected source feeding a balanced Y-connected load. See the figure below.



assuming the *abc* sequence, the phase voltages of a delta-connected source are:

$V_{ab} = V_p \angle 0^0$, $V_{bc} = V_p \angle -120^0$ and $V_{ca} = V_p \angle 120^0$

These are also the line voltages.

The line currents can be obtained in many ways. One way is to apply KVL to loop *aANBba*. In this loop:

$$V_{ab} - V_p \angle 0^0 - I_a Z_Y - I_b Z_Y = Z_Y (I_a - I_b)$$

Or:

$$I_a - I_b = \frac{V_p \angle 0^0}{Z_Y} \qquad (1)$$

But:

$$I_b = I_a \angle -120^0$$

And:

$$I_a - I_b = I_a - I_a \angle -120^0 = I_a \left[1 - \left(\cos 120^0 - j \sin 120^0\right)\right] = I_a \left[1 - \left(-\frac{1}{2} - j\frac{\sqrt{3}}{2}\right)\right]$$

$$= I_a \left[\frac{3}{2} + j\frac{\sqrt{3}}{2}\right] = \sqrt{3} I_a \angle 30^0 \qquad (2)$$

From (1) and (2):

$$\sqrt{3} I_a \angle 30^0 = \frac{V_p \angle 0^0}{Z_Y}$$

Or:

$$I_a = \frac{V_p \angle -30^0}{\sqrt{3} Z_Y}$$

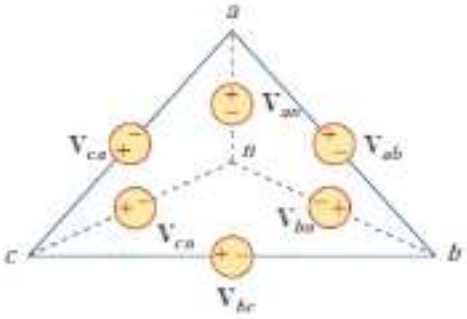
From this, we obtain the other line currents, using the positive phase sequence, i.e..

$$I_b = I_a \angle -120^\circ = \frac{V_P \angle -150^\circ}{\sqrt{3} Z_Y}$$

$$I_C = I_a \angle 120^\circ = \frac{V_P \angle 90^\circ}{\sqrt{3} Z_Y}$$

The phase currents are equal to the line currents.

Another approach to analyze the $\Delta - Y$ circuit is to convert the Δ source into its Y equivalent, as shown below.



Here:

$$V_{an} = \frac{V_{ab} \angle -30^\circ}{\sqrt{3}} = \frac{V_P \angle -30^\circ}{\sqrt{3}}, \qquad V_{bn} = \frac{V_P \angle -150^\circ}{\sqrt{3}} \text{ and } V_{Cn} = \frac{V_P \angle 90^\circ}{\sqrt{3}}$$

Once the source is transformed to Y, the circuit becomes a $Y - Y$ system. Therefore, we can use the equivalent single-phase circuit, as below.



From this circuit:

$$I_a = \frac{V_P \angle -30^\circ}{\sqrt{3} Z_Y}$$

Then:

$$I_b = I_a \angle -120^\circ = \frac{V_P \angle -150^\circ}{\sqrt{3} Z_Y}$$

$$I_C = I_a \angle 120^\circ = \frac{V_P \angle 90^\circ}{\sqrt{3} Z_Y}$$

Which is the same as before.

Alternatively, we may transform the wye-connected load to an equivalent delta-connected load, as before.

Example:

In a balanced $\Delta - Y$ circuit, $V_{ab} = 240 \angle 15^\circ$ and $Z_Y = (12 + j15)\Omega$. Calculate the line current.

$$V_{an} = \frac{V_{ab}}{\sqrt{3}} \angle -30^\circ = \frac{240}{\sqrt{3}} \angle -15^\circ = 138.56 \angle -15^\circ V$$

$$I_a = \frac{V_{an}}{Z_Y} = \frac{138.56 \angle -15^\circ}{12 + j15} = \frac{138.56 \angle -15^\circ}{19.21 \angle 51.34^\circ} = 7.21 \angle -66.34^\circ A$$

$$I_b = 7.21 \angle -186.34^\circ A$$

$$I_c = 7.21 \angle 53.66^\circ A$$

Power in a Balanced Three-Phase System

For a Y-connected load, the phase voltages are:

$v_{AN} = V_m \cos(\omega t)$, $v_{BN} = V_m \cos(\omega t - 120^\circ)$ and $v_{CN} = V_m \cos(\omega t + 120^\circ)$

Assume an inductive load with $Z_Y = Z \angle \theta$. The three phase currents are:

$i_a = I_m \cos(\omega t - \theta)$, $i_b = I_m \cos(\omega t - \theta - 120^\circ)$ and $i_c = I_m \cos(\omega t - \theta + 120^\circ)$

The total instantaneous power in the load is the sum of the instantaneous powers in the three phases; that is,

$$p = p_a + p_b + p_c = v_{AN} i_a + v_{BN} i_b + v_{CN} i_C$$

$$p = V_m I_m [\cos(\omega t)\cos(\omega t - \theta) + \cos(\omega t - 120^\circ)\cos(\omega t - \theta - 120^\circ) + \cos(\omega t + 120^\circ)\cos(\omega t - \theta + 120^\circ)]$$

$$p = \frac{V_m I_m}{2} [\cos(\theta) + \cos(2\omega t - \theta) + \cos(\theta) + \cos(2\omega t - \theta - 240^\circ) + \cos(\theta)\cos(2\omega t - \theta + 240^\circ)]$$

$$p = \frac{V_m I_m}{2} [3\cos(\theta) + \cos(2\omega t - \theta) + \cos(2\omega t - \theta - 240^\circ) + \cos(2\omega t - \theta + 240^\circ)]$$

Let, $2\omega t - \theta = \alpha$, thus:

$$p = \frac{V_m I_m}{2} [3\cos(\theta) + \cos(\alpha) + \cos(\alpha - 240^\circ) + \cos(\alpha + 240^\circ)]$$

$$p = \frac{V_m I_m}{2} [3\cos(\theta) + \cos(\alpha) + \cos(\alpha)\cos 240^\circ + \sin(\alpha)\sin 240^\circ + \cos(\alpha)\cos 240^\circ - \sin(\alpha)\sin 240^\circ]$$

$$p = \frac{V_m I_m}{2} [3\cos(\theta) + \cos(\alpha) + 2\cos(\alpha)\cos 240^\circ]$$

$$p = \frac{V_m I_m}{2} \left[3\cos(\theta) + \cos(\alpha) + 2\cos(\alpha)\left(-\frac{1}{2}\right)\right]$$

$$p = \frac{V_m I_m}{2} [3\cos(\theta)] = 3V_p I_p \cos(\theta) \qquad (1)$$

Where:

$V_p = \frac{V_m}{\sqrt{2}}$ and $I_p = \frac{I_m}{\sqrt{2}}$ are the rms values of the phase voltages and currents.

Equation (1) shows that the total instantaneous power in a balanced three-phase system is constant, it does not change with time as the instantaneous power of each phase does. This result

is true whether the load is Y- or -connected. This is one important reason for using a three-phase system to generate and distribute power.

Since the total instantaneous power is independent of time, the average power per phase P_p for either the Δ-connected load or the Y-connected load is $p/3$, or:

$$P_p = V_p I_p \cos(\theta)$$

and the reactive power per phase is:

$$Q_p = V_p I_p \sin(\theta)$$

The apparent power per phase is:

$$S_p = V_p I_p$$

The complex power per phase is:

$$S_p = P_p + jQ_p = V_p I_p^*$$

The total average power is the sum of the average powers in the phases:

$$P = P_a + P_b + P_c = 3P_p = 3V_p I_p \cos\theta = \sqrt{3} V_L I_L \cos\theta$$

This equation applies for both Y-connected and Δ-connected loads, as for Y-connected loads $I_p = I_L$ and $V_p = \frac{V_L}{\sqrt{3}}$ while for Δ-connected loads $I_p = \frac{I_L}{\sqrt{3}}$ and $V_p = V_L$.

Similarly, the total reactive power is:

$$Q = 3V_p I_p \sin\theta = \sqrt{3} V_L I_L \sin\theta$$

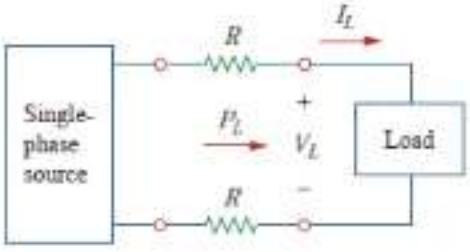
and the total complex power is:

$$S = 3S_p = 3V_p I_p^* = 3I_p^2 Z_p = \frac{3V_p^2}{Z_p^*} = P + jQ = \sqrt{3} V_L I_L \angle\theta$$

A second major advantage of three-phase systems for power distribution is that the three-phase system uses a lesser amount of wire than the single-phase system for the same line voltage and the same absorbed power, P_L.

We will compare a three-phase and a single-phase system supplying the same power using conductors of the same material and with the same length.

For the single-phase system, shown below.

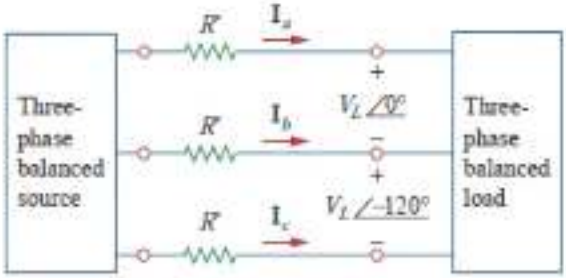


Transmission lines

$I_L = \frac{P_L}{V_L}$ and the power losses are:

$$P_{loss} = 2(I_L)^2 R = 2R\left(\frac{P_L}{V_L}\right)^2 \qquad (1)$$

For the three-phase system shown below.



Transmission lines

$I_L' = \frac{P_L}{\sqrt{3}V_L}$ and the power losses are:

$$P_{loss} = 3(I_L')^2 R' = 3R'\left(\frac{P_L}{\sqrt{3}V_L}\right)^2 = R'\left(\frac{P_L}{V_L}\right)^2 \qquad (2)$$

Equations (1) and (2) shows that for the same power delivered P_L and the same line voltage V_L,

$$\frac{P_{loss}}{P_{Loss}} = \frac{2R\left(\frac{P_L}{V_L}\right)^2}{R'\left(\frac{P_L}{V_L}\right)^2} = \frac{2R}{R'} \qquad (3)$$

But:

$$R = \rho\frac{l}{A} = \rho\frac{l}{\pi r^2} \quad \text{and} \quad R' = \rho\frac{l}{A'} = \rho\frac{l}{\pi r'^2} \qquad (4)$$

In these equations ρ and l are the same for the two systems, thus:

$$\frac{P_{loss}}{P_{Loss}} = \frac{2\rho\frac{l}{\pi r^2}}{\rho\frac{l}{\pi r'^2}} = \frac{2r'^2}{r^2}$$

Here are r and r' are the radiuses of the conductors in the single phase and the three-phase circuits.

If the same power loss is tolerated in both systems, then:

$$\frac{P_{loss}}{P_{Loss}} = \frac{2r'^2}{r^2} = 1$$

Or:

$2r'^2 = r^2$, and:

$$r = \sqrt{2}r' \qquad (5)$$

The ratio of material required is determined by the number of wires and their volumes, so:

$$\frac{material\quad for\quad sin\,gle\quad phase}{material\quad for\quad three\quad phase}=\frac{2\left(\pi r^{2}l\right)}{3\left(\pi r'^{2}l\right)}=\frac{2r^{2}}{3r'^{2}} \qquad (6)$$

Substituting (5) in this equation, we find:

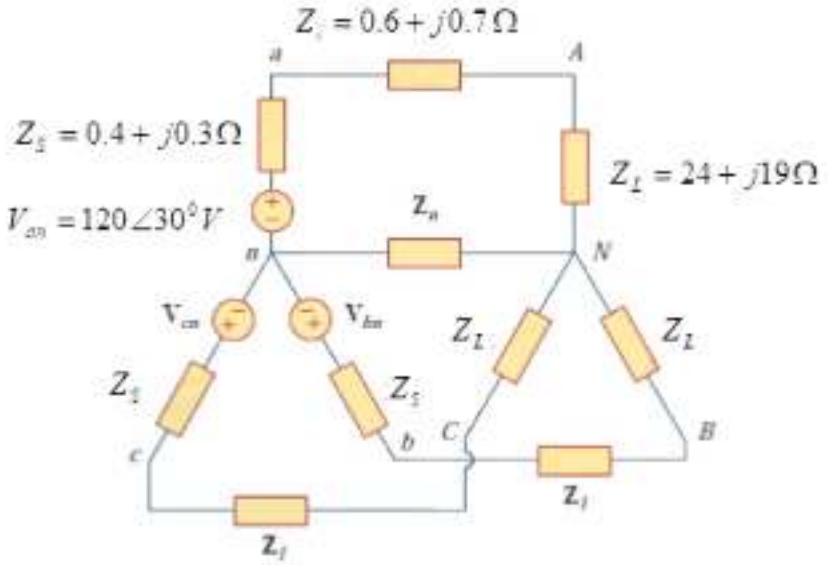
$$\frac{material\quad for\quad sin\,gle\quad phase}{material\quad for\quad three\quad phase}=\frac{2\left(\sqrt{2}r'\right)^{2}}{3r'^{2}}=\frac{4}{3}=1.333$$

Or the single-phase system uses 33 percent more material than the three-phase system or that the three-phase system uses only 75 percent of the material used in the equivalent single-phase system. In other words, considerably less material is needed to deliver the same power with a three-phase system than is required for a single-phase system.

Three-Phase Circuits

Example 1:

For the Y-Y balanced three-phase system shown, assuming a positive sequence for the source voltages and that $V_{an} = 120\angle 30^{0} V$, calculate the complex power at the source and at the load..



Solution:

$$I_a = \frac{V_{an}}{Z_y} = \frac{120\angle 30^{0}}{(0.4 + j0.3) + (0.6 + j0.7) + (24 + j19)} = \frac{120\angle 30^{0}}{32.02\angle 38.66^{0}} = 3.75\angle -8.66^{0} A$$

The complex power generated by the source:

$$S_s = 3V_{an} I_a^{*} = 3(120\angle 30^{0})(3.75\angle 8.66^{0}) = 1350\angle 38.66^{0} = (1054.2 + j843.34) VA$$

The complex power consumed by the load is:

$$S_L = 3|I_{an}|^2 . Z_L = 3(3.75)^2 (24 - j19) = (1011.5 + j800.78) VA$$

The complex power consumed by the line is:

$$S_l = 3|I_{an}|^2 . Z_l = 3(3.75)^2 (0.6 + j0.7) = (25.3 + j29.5) VA$$

The complex power losses inside the source is:

$$S_s = 3|I_{an}|^2 . Z_s = 3(3.75)^2 (0.4 + j0.3) = (17.3 + j13.1) VA$$

Note that $S_s = S_L + S_l + S_{ls}$ which satisfies the power balance. Or, the power generated by the source is equal to the sum of the powers consumed.

Example 2:

Calculate the line current required for a 30-kW three-phase motor having a power factor of 0.85 lagging if it is connected to a balanced source with a line voltage of 440 V.

Solution:

$$P = \sqrt{3} V_L I_L . pf$$

Or:

$$I_L = \frac{P}{\sqrt{3}V_L\, pf} = \frac{30x10^3}{\sqrt{3}(440)(0.85)} = 46.31A$$

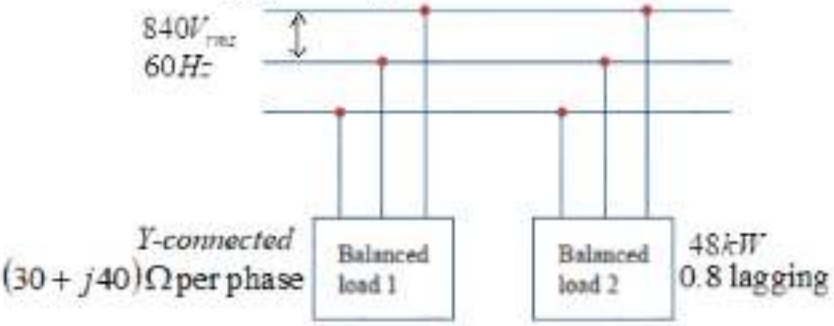
Example 3:

The two balanced loads in the figure below are supplied by an $840V_{rms}$, $60Hz$ line. Load 1 is Y-connected with $(30 - j40)\Omega$ per phase, while load 2 is a balanced three-phase motor drawing $48kW$ at a power factor of 0.8 lagging. Assuming the abc sequence, calculate:

a) the complex power absorbed by the combined load,

b) the kVAR rating of each of the three capacitors -connected in parallel with the load to raise the power factor to unity, and

c) the current drawn from the supply at unity power factor condition.



Solution:

a)

The phase voltage of the supply is:

$$V_{P,rms} = \frac{840}{\sqrt{3}} = 484.97V$$

The current consumed by load 1 is:

$$I_{rms} = \frac{V_{P,rms}}{Z_Y} = \frac{484.97}{30 + j40} = (5.82 - j7.76)A$$

The complex power consumed by load 1 is:

$$S_1 = 3V_{P,rms}\, I_{rms}^* = 3(484.97)(5.82 + j7.6) = (8467.1 + j11290)VA$$

For load 2:

$P_2 = 48kW$, $pf = 0.8$ lagging.

$|S_2| = \frac{P_2}{pf} = \frac{48}{0.8} = 60kVA$ is the apparent power.

But: $|S_2| = \sqrt{P_2^2 + Q_2^2}$, or:

$$Q_2 = \sqrt{|S_2|^2 + P_2^2} = \sqrt{60^2 - 48^2} = 36kVAR$$

Thus, the complex power consumed by load 2 is:

$S_2 = P_2 + jQ_2 = (48 + j36)kVA$

the complex power absorbed by the combined load is:

$$S = S_1 + S_2 = (8467.1 + j11290) + (48 + j36)x10^3 = (56.47 + j47.29)kVA$$

b) at unity power factor, the reactive power consumed by the load and the capacitors is zero, or the generated reactive power by the three capacitors is equal to the consumed reactive power by the load, or:

$Q_C = 47.29kVAR$

Or the kVAR rating of each of the three capacitors is:

$$Q'_C = \frac{47.29}{3} = 15.763kVAR$$

c) at unity power factor, the load consumed only active power, or the active power supplied by the source is:

$P = P_1 + P_2 = 8.47 + 48 = 56.47kW$

the current drawn from the supply is:

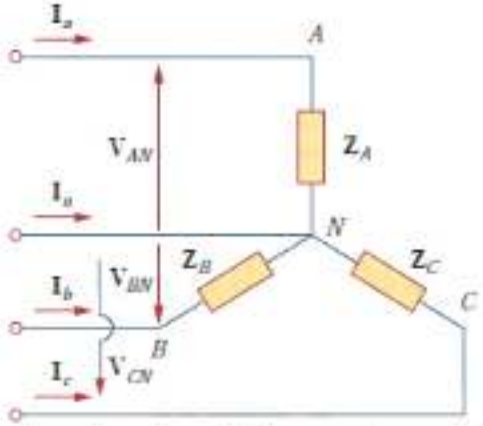
$$I = \frac{P}{\sqrt{3}V_{L,rms}} = \frac{56.47x10^3}{\sqrt{3}(840)} = 38.813A$$

Unbalanced Three-Phase Systems

An unbalanced system is caused by two possible situations:
1) the source voltages are not equal in magnitude and/or differ in phase by angles that are unequal, or
2) load impedances are unequal.

Unbalanced three-phase systems are solved by direct application of mesh and nodal analysis.

The figure below shows an example of an unbalanced three-phase system that consists of balanced source voltages and an unbalanced Y-connected load.



Note that the load is unbalanced, or Z_A, Z_B and Z_C are not equal . The line currents are determined by Ohm's law as:

$$I_a = \frac{V_{AN}}{Z_A}, \quad I_b = \frac{V_{BN}}{Z_B}, \quad I_c = \frac{V_{CN}}{Z_C}$$

This set of unbalanced line currents produces current in the neutral line, or:

$I_n = -(I_a + I_b + I_c)$

In a three-wire system where the neutral line is absent, we can still find the line currents I_a, I_b and I_c using mesh analysis. At node N:

$I_a + I_b + I_c = 0$

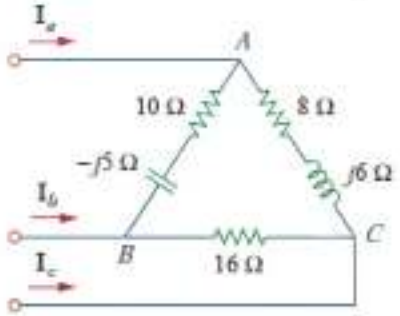
To calculate power in an unbalanced three-phase system we find the power in each phase using:

$P_p = V_p I_p \cos\theta$, $Q_p = V_p I_p \sin\theta$, $S_p = P_p + Q_p = V_p . I_p$

The total power is the sum of the powers in the three phases.

Example 1:

The unbalanced Δ-load in the figure below is supplied by balanced line-to-line voltages of $240V$ in the positive sequence. Find the line currents. Take V_{ab} as reference.



Solution:

The phase currents are:

$$I_{AB} = \frac{V_{AB}}{Z_{AB}} = \frac{240}{10 - j5} = (19.2 + j9.6)A$$

$$I_{BC} = \frac{V_{BC}}{Z_{BC}} = \frac{240\angle -120^\circ}{16} = 15\angle -120^\circ A = (-7.5 - j12.99)$$

$$I_{CA} = \frac{V_{CA}}{Z_{CA}} = \frac{240\angle 120^\circ}{8 + j6} = (2.87 + j23.83)A$$

The line currents are:

$I_a = I_{AB} - I_{CA} = (19.2 + j9.6) - (2.87 + j23.83) = (16.33 - j14.23)A = 21.66\angle -41.07^\circ A$

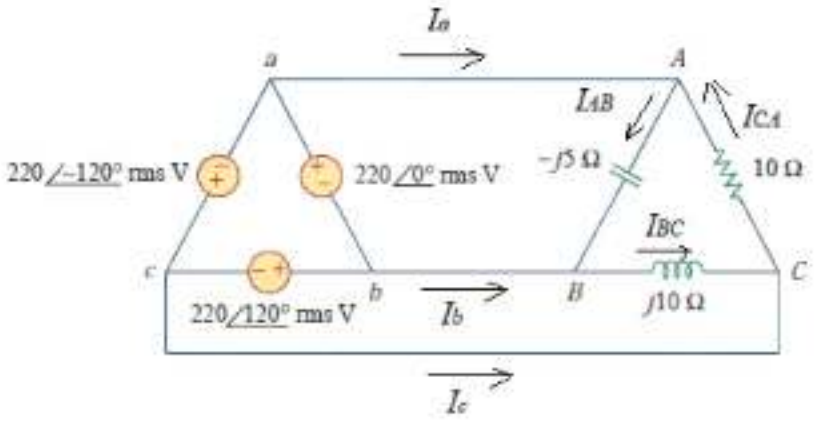
$I_b = I_{BC} - I_{AB} = (-7.5 - j12.99) - (19.2 + j9.6) = (-26.7 - j22.59)A = 34.97\angle 220.23^\circ A$

$I_c = I_{CA} - I_{BC} = (2.87 - j23.83) - (-7.5 - j12.99) = (10.37 + j36.82)A = 38.25\angle 74.27^\circ A$

Note that $I_a + I_b + I_c = 0$

Example 2:

Find the line currents in the unbalanced three-phase circuit of the following figure and the real power absorbed by the load.



Solution:

The phase currents are:

$$I_{AB} = \frac{V_{AB}}{Z_{AB}} = \frac{220}{-j5} = j44A$$

$$I_{BC} = \frac{V_{BC}}{Z_{BC}} = \frac{220\angle 120^{\circ}}{j10} = \frac{220\angle 120^{\circ}}{10\angle 90^{\circ}} = 22\angle 30^{\circ}\,A$$

$$I_{CA} = \frac{V_{BC}}{Z_{BC}} = \frac{220\angle -120^{\circ}}{10} = 22\angle -120^{\circ}\,A$$

The line currents are:

$$I_a = I_{AB} - I_{CA} = j44 - 22\angle -120^{\circ} = 11 + j63.05 = 64\angle 80.1^{\circ}\,A$$

$$I_b = I_{BC} - I_{AB} = 22\angle 30^{\circ} - j44 = 19.05 - j33 = 38.1\angle -60^{\circ}\,A$$

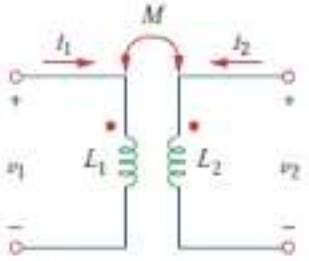
$$I_c = I_{CA} - I_{BC} = -30.1 - j30.1 = 42.5\angle 225^{\circ}\,A$$

Note that:

$$I_a - I_b + I_c = 0$$

Energy in a Coupled Circuit

Consider a circuit with two coupled coils shown below.



As this system is conservative, the energy stored in this system depends on the final state of the system and doesn't depend on how this state has been reached.

Assume that currents i_1 and i_2 are zero initially. so that the energy stored in the coils is zero. If we let i_1 increase from zero to I_1 while maintaining $i_2 = 0$, the power in coil 1 is:

$$p_1(t) = v_1 i_1 = i_1 L_1 \frac{di_1}{dt}$$

and the energy stored in the circuit is:

$$W_1 = \int p_1(t)\,dt = \int_0^{I_1} i_1 L_1 \frac{di_1}{dt}\,dt = \int_0^{I_1} i_1 L_1\,di_1 = \frac{1}{2} L_1 I_1^2$$

If we now maintain $i_1 = I_1$ and increase i_2 from zero to I_2, the mutual voltage induced in coil 1 is $M_{12}\dfrac{di_2}{dt}$, while the mutual voltage induced in coil 2 is zero, since i_1 does not change. The power in the coils is now:

$$p_2(t) = v_1 i_1 + v_2 i_2 = I_1 M_{12} \frac{di_2}{dt} + i_2 L_2 \frac{di_2}{dt}$$

and the energy stored in the circuit is:

$$W_2 = \int p_2(t)\,dt = I_1 M_{12} \int \frac{di_2}{dt}\,dt + L_2 \int i_2 \frac{di_2}{dt}\,dt = I_1 M_{12} \int_0^{I_2} di_2 + L_2 \int_0^{I_2} i_2\,di_2$$

$$= M_{12} I_1 I_2 + \frac{1}{2} L_2 I_2^2$$

The total energy stored in the coils when both i_1 and i_2 have reached constant values is:

$$W = W_1 + W_2 = \frac{1}{2} L_1 I_1^2 + \frac{1}{2} L_2 I_2^2 + M_{12} I_1 I_2 \qquad (1)$$

If we reverse the order by which the currents reach their final values, that is, if we first increase i_2 from zero to I_2 and later increase i_1 from zero to I_1 the total energy stored in the coils is:

$$W = \frac{1}{2} L_1 I_1^2 + \frac{1}{2} L_2 I_2^2 + M_{21} I_1 I_2 \qquad (2)$$

Since the total energy stored should be the same regardless of how we reach the final conditions, thus equating (1) and (2), we conclude that:

$M_{12} = M_{21} = M$

And:

$$W = \frac{1}{2}L_1 I_1^2 + \frac{1}{2}L_2 I_2^2 \pm M I_1 I_2$$

This equation was derived based on the assumption that the coil currents both entered the dotted terminals. If one current enters one dotted terminal while the other current leaves the other dotted terminal, the mutual voltage is negative, so that the mutual energy is also negative. In that case,

$$W = \frac{1}{2}L_1 I_1^2 + \frac{1}{2}L_2 I_2^2 - M I_1 I_2$$

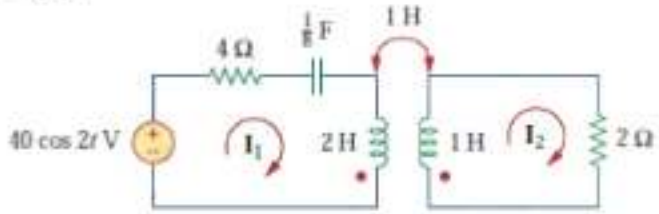
The strength of the magnetic coupling between two magnetically-coupled coils is defined by the coefficient of coupling k, defined as:

$$k = \frac{M}{\sqrt{L_1 L_2}}, \text{ where } 0 \le k \le 1$$

Note that the coupling coefficient is the fraction of the total flux emanating from one coil that links the other coil. If the entire flux produced by one coil links another coil, then $k = 1$ and we have 100 percent coupling, or the coils are said to be perfectly coupled. For $k < 0.5$ coils are said to be loosely coupled; and for $k > 0.5$, they are said to be tightly coupled.

Example:

For the circuit shown below, determine the coupling coefficient and the energy stored in the coupled inductors at $t = 1.5s$



Solution:

$$k = \frac{M}{\sqrt{L_1 L_2}} = \frac{1}{\sqrt{(2)(1)}} = 0.7071$$

In frequency domain:

$40\cos 2t \Rightarrow 40\angle 0^\circ, \qquad \omega = 2rad/s$

$2H \Rightarrow j\omega L_1 = j4\Omega$

$1H \Rightarrow j\omega L_2 = j2\Omega$

$1H \Rightarrow j\omega M = j2\Omega$

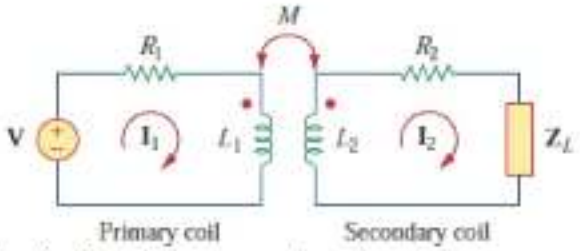
$\frac{1}{8}F \Rightarrow \frac{1}{j\omega C} = -j4\Omega$

Using KVL, for loops 1 and 2:

$$(4 - j4 + j4)I_1 - j2I_2 = 40$$
$$(2 + j2)I_2 - j2I_1 = 0$$
Or:
$$4I_1 - j2I_2 = 40$$
$$(2 + j2)I_2 - j2I_1 = 0$$
From these equations:
$$I_1 = 7.69 + j1.54 = 7.84\angle 11.31^\circ A, \text{ or } i_1(t) = 7.84\cos(2t + 11.31^\circ)$$
$$I_2 = 3.08 + j4.62 = 5.55\angle 56.31^\circ A, \text{ or } i_2(t) = 5.55\cos(2t + 56.31^\circ)$$
At $t = 1.5s$:
$$i_1(1.5) = 7.84\cos\left[2(1.5)\left(\frac{180}{\pi}\right) + 11.31^\circ\right] = -7.83A$$
$$i_2(1.5) = 5.55\cos\left[2(1.5)\left(\frac{180}{\pi}\right) + 56.31^\circ\right] = -3.7A$$
And the total energy is:
$$W = \frac{1}{2}L_1 I_1^2 + \frac{1}{2}L_2 I_2^2 - MI_1 I_2 = \frac{1}{2}(2)(-7.38)^2 + \frac{1}{2}(1)(-3.7)^2 - (1)(-7.38)(-3.7) = 34J$$

Linear Transformers

A transformer has generally two magnetically-coupled coils. See the figure below.



As in the figure, the coil that is directly connected to the voltage source is called the primary winding. The coil connected to the load is called the secondary winding. The resistances R_1 and R_2 are included to account for the losses (power dissipation) in the coils. The transformer is said to be linear if the coils are wound on a magnetically linear material—a material for which the magnetic permeability is constant.

Applying KVL to the two meshes in the figure above,

$$(R_1 + j\omega L_1)I_1 - j\omega M I_2 = V \qquad (1)$$

$$(R_2 + j\omega L_2 + Z_L)I_2 - j\omega M I_1 = 0 \qquad (2)$$

From (2):

$$I_2 = \frac{j\omega M I_1}{R_2 + j\omega L_2 + Z_L} \qquad (3)$$

Substituting (3) in (1):

$$(R_1 + j\omega L_1)I_1 - j\omega M\left(\frac{j\omega M}{R_2 + j\omega L_2 + Z_L}\right)I_1 = V$$

Or:

$$\left[(R_1 + j\omega L_1) + \left(\frac{\omega^2 M^2}{R_2 + j\omega L_2 + Z_L}\right)\right]I_1 = V$$

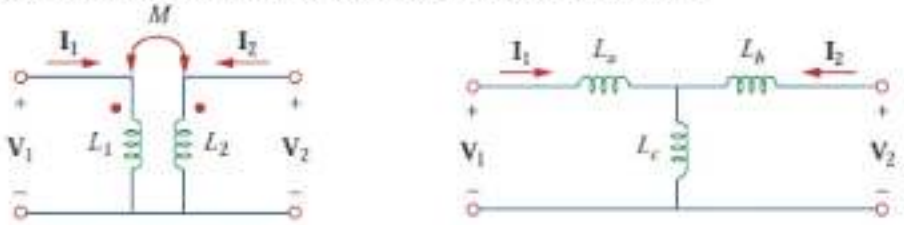
From this the input impedance of the transformer is:

$$Z_{in} = \frac{V}{I_1} = (R_1 + j\omega L_1) + \frac{\omega^2 M^2}{R_2 + j\omega L_2 + Z_L}$$

Notice that the input impedance comprises two terms. The first term is the primary impedance. The second term is due to the coupling between the primary and secondary windings. It is as though this impedance is reflected to the primary. Thus, it is known as the reflected impedance and,

$$Z_R = \frac{\omega^2 M^2}{R_2 + j\omega L_2 + Z_L}$$

It is sometimes convenient to replace a magnetically coupled circuit by an equivalent circuit with no magnetic coupling. We want to replace the linear transformer in the figure below by an equivalent T circuit with no mutual inductance, see the figure below.



For the transformer circuit:

$V_1 = j\omega L_1 I_1 + j\omega M I_2$

$V_2 = j\omega L_2 I_2 + j\omega M I_1$

Or in matrix form:

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} j\omega L_1 & j\omega M \\ j\omega M & j\omega L_2 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} \qquad (4)$$

For the T-equivalent circuit:

$V_1 = j\omega L_a I_1 + j\omega L_c (I_1 + I_2) = j\omega (L_a + L_c) I_1 + j\omega L_c I_2$

$V_2 = j\omega L_b I_2 + j\omega L_c (I_1 + I_2) = j\omega L_c I_1 + j\omega (L_b + L_c) I_2$

Or in matrix form:

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} j\omega (L_a + L_c) & j\omega L_c \\ j\omega L_c & j\omega (L_b + L_c) \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} \qquad (5)$$

Comparing (4) and (5), we find:

$L_a + L_c = L_1, \quad L_c = M, \quad L_b + L_c = L_2$

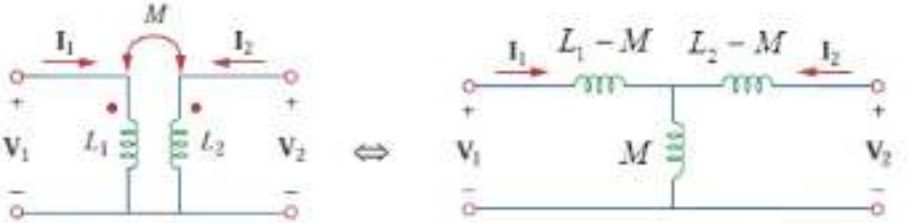
From which:

$L_a + M = L_1, \qquad L_b + M = L_2$

Or:

$L_a = L_1 - M, \qquad L_b = L_2 - M, \quad L_c = M$

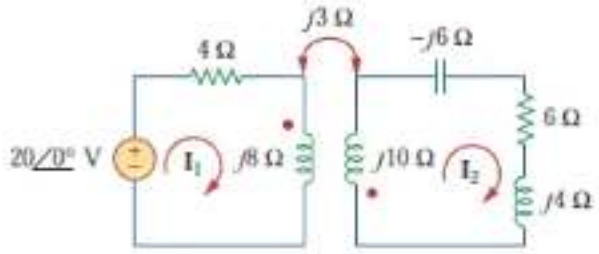
Or, the equivalent circuit is as shown below.



In the same way the Π-equivalent circuit can be found.

Example:

Find the input impedance of the circuit in figure below and the current from the voltage source.



Solution:

From the circuit:

$$(4 - j8)I_1 + j3I_2 = 20\angle 0^0$$

$$(6 + j4 - j6 + j10)I_2 + j3I_1 = 0$$

From this, the current from the voltage source is:

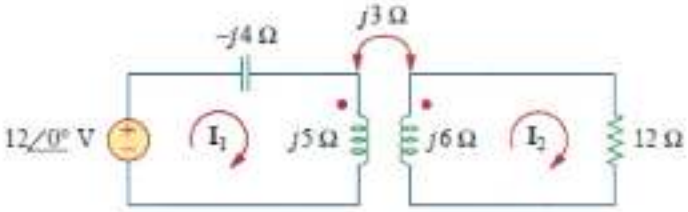
$$I_1 = 1.23 - j1.98 = 2.33\angle -58.05^0\ A$$

The input impedance is:

$$Z_1 = \frac{V}{I_1} = \frac{20\angle 0^0}{2.33\angle -58.05^0} = 8.58\angle 58.05^0\ \Omega$$

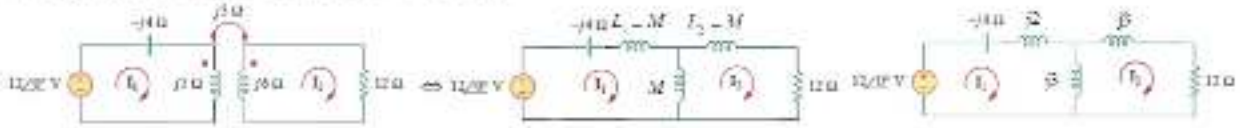
Example:

Using the T-equivalent model, calculate the phasor currents for the magnetically coupled coils, shown below.



Solution:

the equivalent circuit is as shown below.



From the T-equivalent, for loop 1:

$$(-j4 + j2)I_1 + j3(I_1 - I_2) = 12$$

Or:

$$j1.I_1 - j3I_2 = 12 \qquad (1)$$

For loop2:

$$(12 - j3)I_2 + j3(I_2 - I_1) = 0$$

Or:

$$-j3I_1 + (12 + j6)I_2 = 0 \quad (2)$$

From these 2 equations:

$$I_1 = 8.47 - j9.88 = 13.01\angle 49.4^0\, A$$
$$I_2 = 2.82 + j0.706 = 2.91\angle 14.04^0\, A$$

Ideal Transformers

An ideal transformer is one with perfect coupling ($k = 1$). It consists of two (or more) coils with a large number of turns wound on a common core of high permeability. Because of this high permeability of the core, the flux links all the turns of both coils, thereby resulting in a perfect coupling.

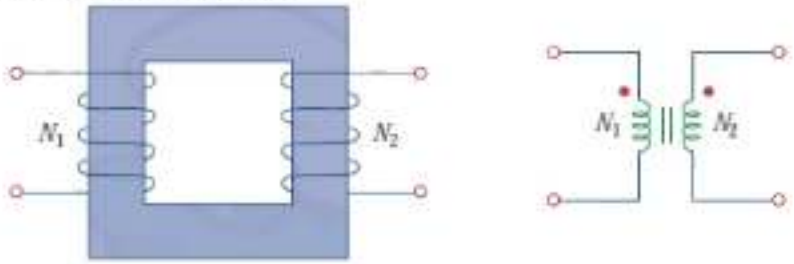
In general, A transformer is said to be ideal if it has the following properties:

1) Coils have very large inductances, or $L_1 \to \infty$, $L_2 \to \infty$ and $M \to \infty$

2) Coupling coefficient is equal to unity, or $k = 1$

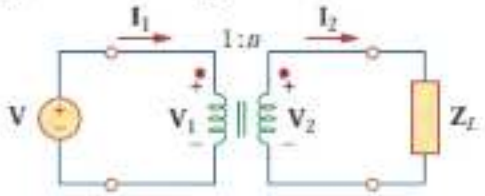
3) Primary and secondary coils are lossless, or $R_1 = 0$ and $R_2 = 0$.

Iron-core transformers, as shown below, are close approximations to ideal transformers. These are used in power systems and electronics.



The vertical lines between the coils indicate an iron core as distinct from the air core used in linear transformers. The primary winding has N_1 turns; the secondary winding has N_2 turns.

When a sinusoidal voltage is applied to the primary winding, as shown in the figure below, the same magnetic flux goes through both windings.



According to Faraday's law, the voltage across the primary and secondary windings are:

$$v_1 = N_1 \frac{d\phi}{dt}$$

$$v_2 = N_2 \frac{d\phi}{dt}$$

Thus:

$$\frac{v_2}{v_1} = \frac{N_2}{N_1} = n$$

where n is the turns ratio or transformation ratio. We can use the phasor voltages and rather than the instantaneous values, thus:

$$\frac{V_2}{V_1} = \frac{N_2}{N_1} = n$$

As the ideal transformer has no power losses, thus the power supplied to the primary must equal the power absorbed by the secondary, or:

$$v_1 i_1 = v_2 i_2$$

Or:

$$\frac{v_2}{v_1} = \frac{i_1}{i_2} = n$$

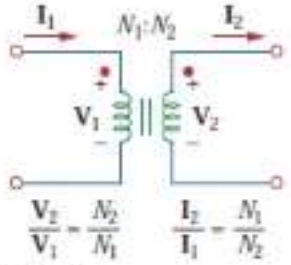
Or, in phasors:

$$\frac{V_2}{V_1} = \frac{I_1}{I_2} = n$$

Or, the primary and secondary currents are related to the turns ratio in the inverse manner as the voltages.

Note that when $n > 1$, $V_2 > V_1$ and the transformer is a step-up transformer. While when $n < 1$, $V_2 < V_1$ and the transformer is a step-down transformer. And when $n = 1$, $V_2 = V_1$ and the transformer is only an isolation transformer.

For polarity of the currents and the voltages in the primary and the secondary windings, note that the primary current enters the primary winding (as it is supplied from the source), while the secondary current leaves the secondary winding as it is supplying the load, see the figure below.



The complex power in the primary winding is:

$$S_1 = V_1 \cdot I_1^* = \frac{V_2}{n} \cdot n I_2^* = V_2 \cdot I_2^* = S_2$$

Or, the complex power supplied to the primary is delivered to the secondary without loss. The transformer absorbs no power. Of course, we should expect this, since the ideal transformer is lossless.

The input impedance as seen by the source is:

$$Z_1 = \frac{V_1}{I_1} = \frac{\frac{V_2}{n}}{n I_2} = \frac{1}{n^2}\frac{V_2}{I_2} = \frac{1}{n^2} Z_2$$

Note that Z_2 is the load impedance, or:

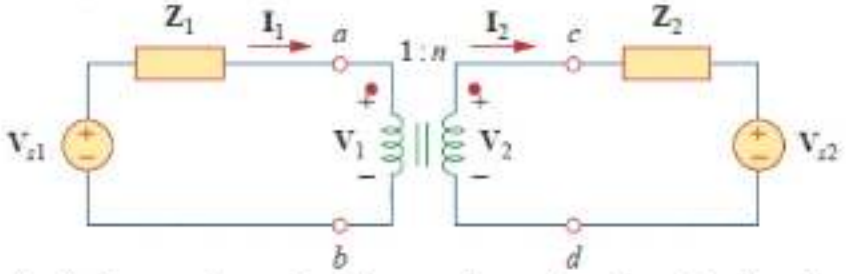
$$Z_2 = Z_L$$

Thus:

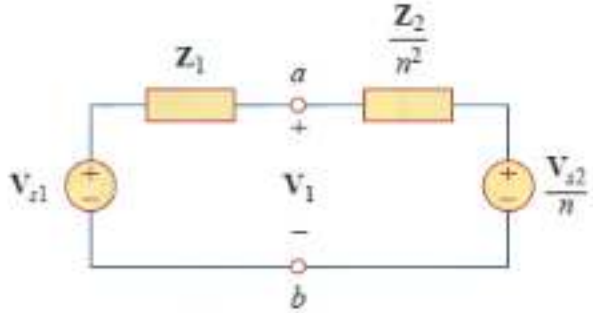
$$Z_1 = \frac{1}{n^2} Z_L$$

The input impedance is also called the *reflected impedance*, since it appears as if the load impedance is reflected to the primary side. This ability of the transformer to transform a given impedance into another impedance provides us a means of *impedance matching* to ensure maximum power transfer.

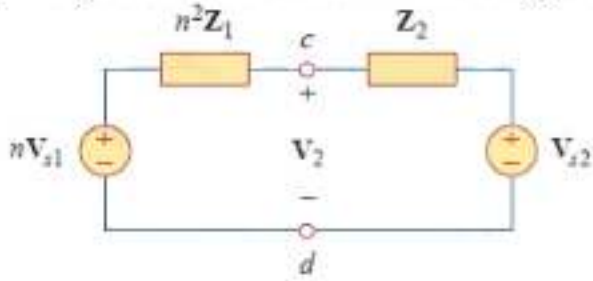
In analyzing a circuit containing an ideal transformer, it is common practice to eliminate the transformer by reflecting impedances and sources from one side of the transformer to the other. Consider the following circuit



To analyze this circuit the secondary side f the transformer is reflected to the primary, as shown below.



We can also reflect the primary side of the circuit to the secondary, as shown in the figure below.



Example:
The primary current to an ideal transformer rated at 3300/110V is 5 A. Calculate:
a) the turns ratio.
b) the kVA rating,
c) the secondary current

Solution:

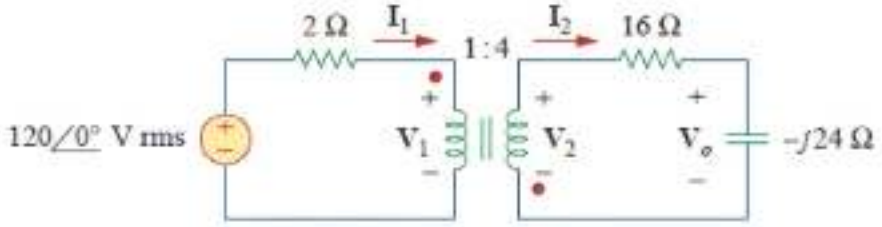
a) $n = \dfrac{V_2}{V_1} = \dfrac{110}{3300} = 1/30$

b) $S = V_1 I_1 = V_2 I_2 = 3300x5 = 16.5kVA$

c) $I_2 = \frac{S}{V_2} = \frac{16500}{110} = 150A$

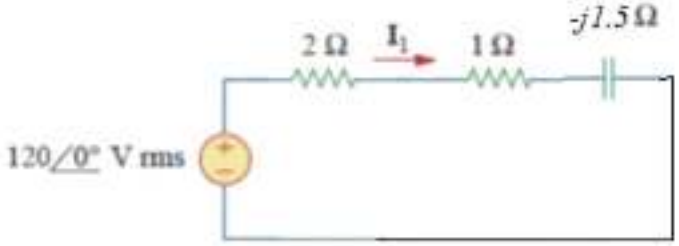
Example:

In the ideal transformer circuit of the figure below, find V_o and the complex power supplied by the source.



Solution:

Reflecting the impedance of the secondary side to the primary, the equivalent circuit becomes:



Note that the secondary impedance as referred to primary is:

$$Z_2' = \frac{Z_2}{n^2} = \frac{16 - j24}{4^2} = 1 - j1.5 \ \ \Omega$$

From this circuit:

$$I_1 = \frac{V_s}{Z} = \frac{120\angle 0^0}{3 - j1.5} = 32 + j16 \ \ A$$

$$I_2 = -\frac{I_1}{n} = -\frac{32 + j16}{4} = -8 - j4 \ \ A$$

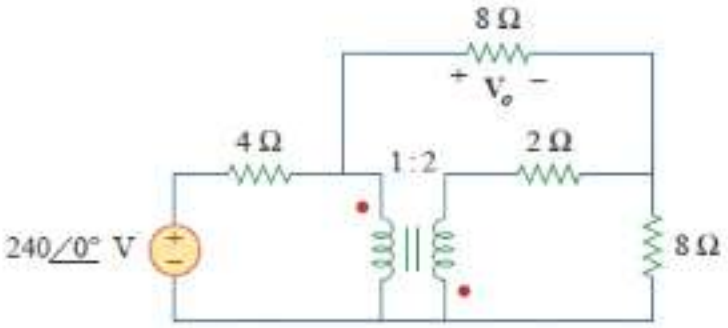
$$V_o = -j24(-8 - j4) = -96 + j192 = 2147\angle 116.6^0 V$$

And

$$S_1 = V_1 I_1 = 120\angle 0^0 (32 + j16) = 3840 + j1920 = 4.294\angle 26.57^0 kVA$$

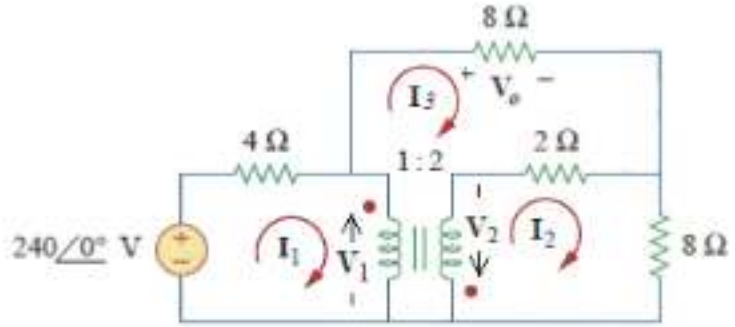
Example:

Find V_o in the circuit below.



Solution:

Reflection to the secondary or primary side cannot be done with this circuit because there is a direct connection between the primary and the secondary sides due to the 8Ω resistor. We apply mesh analysis.



For mesh 1:

$$4I_1 + V_1 = 240 \qquad (1)$$

$$8I_3 + 2(I_3 - I_2) + V_2 = 0 \qquad (2)$$

$$8I_3 + 2(I_3 - I_2) - V_1 - V_2 = 0 \qquad (3)$$

From the relation: $\frac{V_1}{V_2} = \frac{1}{2}$

$$V_2 = 2V_1 \qquad (4)$$

And from the relation $\frac{i_1 - i_2}{i_2} = \frac{2}{1}$, we have:

$$I_1 - I_2 = -2I_1 \qquad (5)$$

From these equations, we find:

$I_1 = 140A$

$I_2 = -60A$

$I_3 = 12A$

$V_1 = -320A$

$V_2 = 640V$

And:

$V_0 = 8I_3 = 96V$